Moulay Ismail University
École Normale Supérieure (ENS–Meknès)
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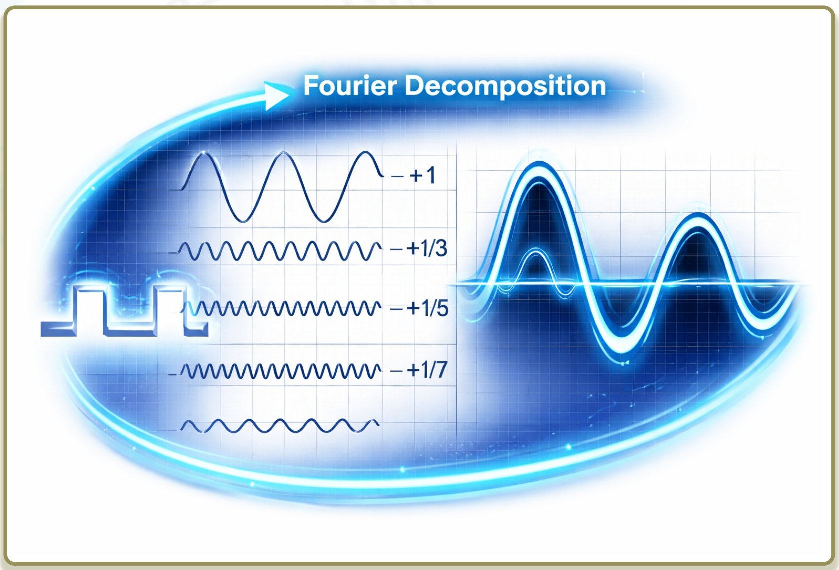
Applied Harmonic Analysis



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Chapter 1

The Fourier Transform on $\mathbb{R}$: Theory and Applications

The theory of Fourier analysis originates in the work of Joseph Fourier (1768–1830), who introduced the idea of representing functions as superpositions of trigonometric waves in his study of heat propagation.
Since then, Fourier analysis has become a central tool in mathematics, physics, and engineering, providing a bridge between time (or space) and frequency representations.
In this chapter, we begin with Fourier series on the torus, which provide a discrete frequency decomposition of periodic functions. We then explain how this theory naturally leads to the Fourier transform on $\mathbb{R}$, where the frequency variable becomes continuous.

The development of the Fourier transform theory will follow a progressive approach, moving from intuitive constructions to fully rigorous functional frameworks. More precisely, we proceed through the following stages :

1. **Periodic functions (Fourier series on $\mathbb{T}$).**
   We start with Fourier series on the torus
   $$\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z},$$
   where functions are decomposed into discrete frequencies. This provides the fundamental spectral viewpoint and serves as a prototype for the continuous theory.

2. **First insight : compactly supported functions and periodization.**
   We then study compactly supported functions and construct periodic functions via periodization. This step provides an intuitive bridge between Fourier series and Fourier transform, and suggests the transition from discrete to continuous frequencies.

3. **Functions with moderate growth.**
   We introduce a first definition of the Fourier transform using improper integrals for functions with moderate growth. This stage is mainly heuristic and highlights the necessity of decay conditions.

4. **Schwartz space $\mathcal{S}(\mathbb{R})$.**
   We then develop the theory on the Schwartz space, where the Fourier transform is well-defined, stable under differentiation, and satisfies the inversion formula. This constitutes the first fully rigorous framework.

5. **Fourier transform on $L^1(\mathbb{R})$.**
   The transform is extended to integrable functions, where it is defined by an absolutely convergent integral and yields bounded continuous functions.

6. **Fourier transform on $L^2(\mathbb{R})$.**
   Using the density of $\mathcal{S}(\mathbb{R})$ and Plancherel's theorem, the Fourier transform is extended to a unitary operator on $L^2(\mathbb{R})$.

7. **Compatibility on $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.**
   We verify that the L^1 and L^2 definitions of the Fourier transform coincide almost everywhere on their intersection.

8. **Tempered distributions.**
   Finally, the Fourier transform is extended to tempered distributions, allowing the treatment of generalized functions and completing the functional analytic framework.

I - Fourier Series on the Torus

Before introducing the Fourier transform on $\mathbb{R}$, let us recall the basic facts about Fourier series on the circle. This discrete theory will serve as a model for the continuous Fourier transform.

1) The One-Dimensional Torus $\mathbb{T}$

Definition 1.1

The one-dimensional torus is the quotient space

$$\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}.$$

Two real numbers $\theta, \theta' \in \mathbb{R}$ represent the same point of $\mathbb{T}$ if and only if

$$\theta - \theta' \in 2\pi\mathbb{Z}.$$

Thus, the point represented by θ is the equivalence class

$$[\theta] = \{\theta + 2\pi k : \ k \in \mathbb{Z}\}.$$

Geometrically, $\mathbb{T}$ is obtained by wrapping the real line around the unit circle and identifying θ with $\theta + 2\pi$.

This identification is realized by the canonical map

$$\begin{aligned} \Phi : \quad \mathbb{R}/2\pi\mathbb{Z} &\longrightarrow \quad \mathbb{S}^1 \\ [\theta] &\longmapsto \quad e^{i\theta} \end{aligned}$$

This map is well-defined and establishes a bijection between $\mathbb{T}$ and the unit circle

$$\mathbb{S}^1 = \{e^{i\theta} : \ \theta \in \mathbb{R}\}.$$

Consequently, one can identify $\mathbb{T}$ with $\mathbb{S}^1$, or equivalently with any interval of length 2π, for instance $[-\pi, \pi)$, where the endpoints are identified.

Proposition 1.2

The map Φ is well-defined and is an isomorphism of topological groups.

Proof:
If $\theta' = \theta + 2\pi k$ for some $k \in \mathbb{Z}$, then

$$e^{i\theta'} = e^{i(\theta+2\pi k)} = e^{i\theta} e^{2\pi i k} = e^{i\theta},$$

so Φ is well-defined.

Moreover, Φ is a group homomorphism. Indeed, for all $\theta, \varphi \in \mathbb{R}$,

$$\Phi([\theta + \varphi]) = e^{i(\theta+\varphi)} = e^{i\theta} e^{i\varphi} = \Phi([\theta])\Phi([\varphi]).$$

The map Φ is surjective, since every point of $\mathbb{S}^1$ can be written as $e^{i\theta}$ for some $\theta \in \mathbb{R}$.

It is injective, since

$$e^{i\theta} = e^{i\varphi} \quad \Longrightarrow \quad \theta - \varphi \in 2\pi\mathbb{Z},$$

which means that $[\theta] = [\varphi]$ in $\mathbb{T}$.

Hence Φ is bijective.

Let $\pi : \mathbb{R} \to \mathbb{T}$ be the canonical projection, $\pi(\theta) = [\theta]$, and define

$$f : \mathbb{R} \to \mathbb{S}^1, \qquad f(\theta) = e^{i\theta}.$$

Since $f(\theta + 2\pi k) = f(\theta)$ for every $k \in \mathbb{Z}$, the map f is constant on equivalence classes modulo 2π. Therefore, by the universal property of the quotient, there exists a unique map

$$\Phi : \mathbb{T} \to \mathbb{S}^1$$

such that the following diagram commutes :

$$\begin{tikzcd}
\mathbb{R} \arrow[rr, "f"] \arrow[dr, "\pi"'] & & \mathbb{S}^1 \\
& \mathbb{T} \arrow[ur, "\Phi"'] &
\end{tikzcd}$$

that is,

$$f = \Phi \circ \pi.$$

Now the map $f(\theta) = e^{i\theta}$ is continuous on $\mathbb{R}$. Since π is the quotient map and $f = \Phi \circ \pi$, it follows from the definition of the quotient topology that Φ is continuous.

Finally, since $\mathbb{T}$ is compact and $\mathbb{S}^1$ is Hausdorff, every continuous bijection from $\mathbb{T}$ onto $\mathbb{S}^1$ is a homeomorphism. Therefore, Φ^{-1} is also continuous. ■

Definition 1.3

A function on $\mathbb{T}$ is identified with a 2π-periodic function on $\mathbb{R}$, that is, a function $f : \mathbb{R} \to \mathbb{C}$ such that

$$f(\theta + 2\pi) = f(\theta), \qquad \forall \theta \in \mathbb{R}.$$

Integration on $\mathbb{T}$ is defined by

$$\int_{\mathbb{T}} f(\theta)\, d\theta := \int_{-\pi}^{\pi} f(\theta)\, \frac{d\theta}{2\pi}.$$

2) $L^p(\mathbb{T})$ Spaces

Definition 1.4

For $1 \leqslant p < \infty$, we define

$$L^p(\mathbb{T}) = \left\{ f \text{ measurable and } 2\pi\text{-periodic} : \int_{-\pi}^{\pi} |f(\theta)|^p \frac{d\theta}{2\pi} < \infty \right\}.$$

Also,

$$L^\infty(\mathbb{T}) = \left\{ f \text{ measurable and } 2\pi\text{-periodic} : \operatorname*{ess\,sup}_{\theta \in [-\pi,\pi]} |f(\theta)| < \infty \right\}.$$

We recall that, for a measurable function f on $[-\pi, \pi]$, the *essential supremum* of $|f|$ is defined by

$$\operatorname*{ess\,sup}_{\theta \in [-\pi,\pi]} |f(\theta)| := \inf \{ M \geqslant 0 : \ |f(\theta)| \leqslant M \text{ for almost every } \theta \in [-\pi, \pi] \}.$$

In other words, $\operatorname{ess\,sup} |f|$ is the smallest constant M such that

$$|f(\theta)| \leqslant M \quad \text{outside a set of measure zero.}$$

In particular, $L^2(\mathbb{T})$ is a Hilbert space, which makes it especially suitable for Fourier analysis, with inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(\theta)\overline{g(\theta)} \frac{d\theta}{2\pi}.$$

3) The Exponential System

Let us now introduce the fundamental building blocks of the theory : complex exponentials.

Definition 1.5

For each $k \in \mathbb{Z}$, define

$$e_k(\theta) := e^{ik\theta}.$$

These functions represent pure oscillations of frequency k. The following orthogonality property is the key algebraic ingredient that allows us to decompose functions into independent frequency components.

Proposition 1.6: Orthogonality relations

The family $(e_k)_{k \in \mathbb{Z}}$ is orthonormal in $L^2(\mathbb{T})$, that is,

$$\langle e_k, e_m \rangle = \delta_{k,m} = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases}.$$

Equivalently, for all $m, n \in \mathbb{Z}$, one has

$$\int_{-\pi}^{\pi} e^{im\theta}\overline{e^{in\theta}} \frac{d\theta}{2\pi} = \delta_{m,n}.$$

Proof:
We compute

$$\langle e_k, e_m \rangle = \int_{-\pi}^{\pi} e^{ik\theta}\overline{e^{im\theta}}\,\frac{d\theta}{2\pi} = \int_{-\pi}^{\pi} e^{i(k-m)\theta}\,\frac{d\theta}{2\pi}.$$

If $k = m$, then the integrand is identically equal to 1, and the integral equals 1.
If $k \neq m$, then

$$\int_{-\pi}^{\pi} e^{i(k-m)\theta}\,\frac{d\theta}{2\pi} = \frac{1}{2\pi}\left[\frac{e^{i(k-m)\theta}}{i(k-m)}\right]_{-\pi}^{\pi}.$$

Since

$$e^{i(k-m)\pi} = e^{-i(k-m)\pi} = (-1)^{k-m},$$

the two boundary terms cancel, and the integral is equal to 0. ■

4) Fourier Coefficients

We are now ready to quantify how much each frequency contributes to a function.

Definition 1.7

Let $f \in L^1(\mathbb{T})$. For each $k \in \mathbb{Z}$, the k-th Fourier coefficient of f is defined by

$$\widehat{f}(k) = \int_{-\pi}^{\pi} f(\theta)e^{-ik\theta}\,\frac{d\theta}{2\pi}.$$

If $f \in L^2(\mathbb{T})$, then

$$\boxed{\widehat{f}(k) = \langle f, e_k \rangle}$$

These coefficients measure the projection of f onto each oscillation e_k.

Proposition 1.8

Let $f \in L^1(\mathbb{T})$ and $k \in \mathbb{Z}$. Then the Fourier coefficient

$$\widehat{f}(k) = \int_{-\pi}^{\pi} f(\theta)e^{-ik\theta}\,\frac{d\theta}{2\pi}$$

is well-defined.

Proof:
Since $|e^{-ik\theta}| = 1$, we have

$$|f(\theta)e^{-ik\theta}| = |f(\theta)|.$$

Thus

$$\int_{-\pi}^{\pi} |f(\theta)e^{-ik\theta}|\,\frac{d\theta}{2\pi} = \int_{-\pi}^{\pi} |f(\theta)|\,\frac{d\theta}{2\pi} < \infty.$$

Hence the integral is absolutely convergent.

Moreover, if $f = g$ almost everywhere, then

$$f(\theta)e^{-ik\theta} = g(\theta)e^{-ik\theta} \quad \text{a.e.},$$

which implies equality of the integrals. Thus $\widehat{f}(k)$ depends only on the class of f in $L^1(\mathbb{T})$. ■

Proposition 1.9: Extraction of Fourier coefficients

Let $f \in L^1(\mathbb{T})$ and assume that

$$f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\theta) + \sum_{n=1}^{\infty} b_n \sin(n\theta),$$

with convergence (for instance in $L^2(\mathbb{T})$). Then for every $m \geqslant 1$,

$$a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(m\theta)\,d\theta, \qquad b_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(m\theta)\,d\theta,$$

and

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\,d\theta.$$

Proof:
Multiply the series by $\cos(m\theta)$ and integrate over $[-\pi,\pi]$. Using the orthogonality relations :

$$\int_{-\pi}^{\pi} \cos(n\theta)\cos(m\theta)d\theta = \begin{cases} \pi, & n = m, \\ 0, & n \neq m, \end{cases}$$

and

$$\int_{-\pi}^{\pi} \sin(n\theta)\cos(m\theta)d\theta = 0,$$

all terms vanish except the one corresponding to $n = m$, yielding

$$\int_{-\pi}^{\pi} f(\theta)\cos(m\theta)d\theta = a_m\pi.$$

The same argument applies for b_m using $\sin(m\theta)$, and for a_0 using the constant function. ■

Example 1 : (Step Function)

Consider the 2π-periodic function defined on $[0, 2\pi]$ by

$$f(t) = \begin{cases} 1, & 0 < t < \pi, \\ 0, & \pi < t < 2\pi. \end{cases}$$

We compute its Fourier coefficients.
First,

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(t)\,dt = \frac{1}{\pi}\int_0^{\pi} 1\,dt = 1.$$

For $m \geqslant 1$,

$$a_m = \frac{1}{\pi}\int_0^{2\pi} f(t)\cos(mt)\,dt = \frac{1}{\pi}\int_0^{\pi}\cos(mt)\,dt = \frac{1}{\pi}\left[\frac{\sin(mt)}{m}\right]_0^{\pi} = 0.$$

Similarly,

$$b_m = \frac{1}{\pi}\int_0^{2\pi} f(t)\sin(mt)\,dt = \frac{1}{\pi}\int_0^{\pi}\sin(mt)\,dt = \frac{1}{\pi}\left[-\frac{\cos(mt)}{m}\right]_0^{\pi}.$$

Hence,

$$b_m = \frac{1}{\pi m}\left(1 - \cos(m\pi)\right) = \frac{1}{\pi m}\left(1 - (-1)^m\right).$$

Therefore,

$$b_m = \begin{cases} 0, & m \text{ even}, \\ \dfrac{2}{\pi m}, & m \text{ odd}. \end{cases}$$

The Fourier series of f is thus

$$f(t) = \frac{1}{2} + \sum_{\substack{m \geqslant 1 \\ m \text{ odd}}} \frac{2}{\pi m} \sin(mt).$$

At points of discontinuity (such as $t = \pi$), the series converges to the midpoint value $\frac{1}{2}$.

5) Fourier Series

We can now reconstruct a function from its Fourier coefficients.

Definition 1.10

The Fourier series of f is the formal expansion

$$f(\theta) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\theta) + \sum_{n=1}^{\infty} b_n \sin(n\theta)$$

or, equivalently, in complex form,

$$f(\theta) \sim \sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ik\theta}.$$

Its N-th partial sum is

$$S_N f(\theta) := \sum_{|k| \leqslant N} \widehat{f}(k) e^{ik\theta}.$$

The real and complex forms of the Fourier series contain exactly the same information ; the next proposition makes the correspondence explicit.

Proposition 1.11

Assume that the Fourier series of f is written in the real form

$$f(\theta) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\theta) + \sum_{n=1}^{\infty} b_n \sin(n\theta),$$

and in the complex form

$$f(\theta) \sim \sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ik\theta}.$$

Then

$$\widehat{f}(0) = \frac{a_0}{2},$$

and for every $n \geqslant 1$,

$$\widehat{f}(n) = \frac{1}{2}(a_n - ib_n), \qquad \widehat{f}(-n) = \frac{1}{2}(a_n + ib_n).$$

Proof:
Using Euler's identities

$$\cos(n\theta) = \frac{e^{in\theta} + e^{-in\theta}}{2}, \qquad \sin(n\theta) = \frac{e^{in\theta} - e^{-in\theta}}{2i},$$

we rewrite the real Fourier expansion as

$$f(\theta) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \frac{e^{in\theta} + e^{-in\theta}}{2} + \sum_{n=1}^{\infty} b_n \frac{e^{in\theta} - e^{-in\theta}}{2i}.$$

Group together the coefficients of $e^{in\theta}$ and $e^{-in\theta}$. We obtain

$$f(\theta) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} + \frac{b_n}{2i}\right) e^{in\theta} + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} - \frac{b_n}{2i}\right) e^{-in\theta},$$

this becomes

$$f(\theta) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \frac{1}{2}(a_n - ib_n)e^{in\theta} + \sum_{n=1}^{\infty} \frac{1}{2}(a_n + ib_n)e^{-in\theta}.$$

Comparing with the complex Fourier expansion

$$f(\theta) \sim \sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ik\theta},$$

we deduce

$$\widehat{f}(0) = \frac{a_0}{2}, \qquad \widehat{f}(n) = \frac{1}{2}(a_n - ib_n), \qquad \widehat{f}(-n) = \frac{1}{2}(a_n + ib_n) \quad (n \geqslant 1).$$

This completes the proof. ■

Proposition 1.12

For every trigonometric polynomial

$$P(\theta) = \sum_{|k|\leqslant N} c_k e^{ik\theta},$$

one has

$$\widehat{P}(k) = c_k, \qquad |k| \leqslant N,$$

and

$$\widehat{P}(k) = 0, \qquad |k| > N.$$

Proof:
Using linearity and orthogonality,

$$\widehat{P}(m) = \int_{-\pi}^{\pi} \left(\sum_{|k|\leqslant N} c_k e^{ik\theta}\right) e^{-im\theta}\,\frac{d\theta}{2\pi} = \sum_{|k|\leqslant N} c_k\langle e_k, e_m\rangle.$$

Hence $\widehat{P}(m) = c_m$ if $|m| \leqslant N$, and $\widehat{P}(m) = 0$ otherwise. ■

Theorem 1.13: Fourier Expansion in $L^2(\mathbb{T})$

For every $f \in L^2(\mathbb{T})$,
$$S_N f \longrightarrow f \qquad \text{in } L^2(\mathbb{T}),$$
where
$$S_N f(\theta) := \sum_{|k| \leqslant N} \widehat{f}(k) e^{ik\theta}.$$

Equivalently,
$$\lim_{N\to\infty} \int_{-\pi}^{\pi} \left| f(\theta) - \sum_{|k|\leqslant N} \widehat{f}(k) e^{ik\theta} \right|^2 \frac{d\theta}{2\pi} = 0.$$

Moreover, Parseval's identity holds :
$$\|f\|_{L^2(\mathbb{T})}^2 = \sum_{k\in\mathbb{Z}} |\widehat{f}(k)|^2.$$

Proof:
We already know that the family
$$(e_k)_{k\in\mathbb{Z}}, \qquad e_k(\theta) := e^{ik\theta},$$
is orthonormal in $L^2(\mathbb{T})$.

We first show that trigonometric polynomials are dense in $L^2(\mathbb{T})$. Recall that
$$C^0(\mathbb{T}) := \{h : \mathbb{T} \to \mathbb{C}\ ;\ h \text{ is continuous}\}$$
denotes the space of continuous 2π-periodic functions on $\mathbb{T}$, endowed with the norm
$$\|h\|_{C^0(\mathbb{T})} := \max_{|\theta|\leqslant\pi} |h(\theta)|.$$

For every $h \in C^0(\mathbb{T})$, we have
$$\|h\|_{L^2(\mathbb{T})} = \left(\int_{-\pi}^{\pi} |h(\theta)|^2 \frac{d\theta}{2\pi}\right)^{1/2} \leqslant \left(\|h\|_{C^0(\mathbb{T})}^2 \int_{-\pi}^{\pi} \frac{d\theta}{2\pi}\right)^{1/2} = \|h\|_{C^0(\mathbb{T})}.$$

Thus $C^0(\mathbb{T})$ is continuously embedded into $L^2(\mathbb{T})$.

Now we use two classical density facts :

1. $C^0(\mathbb{T})$ is dense in $L^2(\mathbb{T})$;
2. trigonometric polynomials are dense in $C^0(\mathbb{T})$.

Let $f \in L^2(\mathbb{T})$ and let $\varepsilon > 0$. By density of $C^0(\mathbb{T})$ in $L^2(\mathbb{T})$, there exists $h \in C^0(\mathbb{T})$ such that
$$\|f - h\|_{L^2(\mathbb{T})} \leqslant \varepsilon.$$

By density of trigonometric polynomials in $C^0(\mathbb{T})$, there exists a trigonometric polynomial
$$P(\theta) = \sum_{\text{finite}} a_k e^{ik\theta}$$
such that
$$\|h - P\|_{C^0(\mathbb{T})} \leqslant \varepsilon.$$
Hence
$$\|h - P\|_{L^2(\mathbb{T})} \leqslant \|h - P\|_{C^0(\mathbb{T})} \leqslant \varepsilon.$$

Therefore,

$$\|f - P\|_{L^2(\mathbb{T})} \leqslant \|f - h\|_{L^2(\mathbb{T})} + \|h - P\|_{L^2(\mathbb{T})} \leqslant 2\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, trigonometric polynomials are dense in $L^2(\mathbb{T})$.

For each $N \geqslant 0$, let

$$E_N := \operatorname{span}\{e_k : \ |k| \leqslant N\}.$$

We now show that

$$S_N f = \sum_{|k|\leqslant N} \widehat{f}(k) e_k$$

is the orthogonal projection of f onto E_N.
Indeed, let

$$P_N(\theta) = \sum_{|k|\leqslant N} a_k e_k(\theta) \in E_N.$$

For every m with $|m| \leqslant N$, orthonormality gives

$$\langle P_N, e_m \rangle = \sum_{|k|\leqslant N} a_k \langle e_k, e_m \rangle = a_m.$$

Thus

$$\langle f - P_N, e_m \rangle = \langle f, e_m \rangle - \langle P_N, e_m \rangle = \widehat{f}(m) - a_m.$$

Hence the condition $f - P_N \perp E_N$ is equivalent to

$$a_m = \widehat{f}(m) \qquad \text{for all } |m| \leqslant N.$$

Therefore the unique element of E_N orthogonal to the error is

$$S_N f = \sum_{|k|\leqslant N} \widehat{f}(k) e_k.$$

In particular,

$$f - S_N f \perp E_N.$$

Since

$$\bigcup_{N\geqslant 0} E_N$$

is exactly the set of trigonometric polynomials, which is dense in $L^2(\mathbb{T})$, it follows that the orthogonal projections $S_N f$ converge to f in $L^2(\mathbb{T})$:

$$S_N f \longrightarrow f \qquad \text{in } L^2(\mathbb{T}).$$

This proves the first assertion of the theorem.

We now prove Parseval's identity. Since $S_N f \in E_N$ and $f - S_N f \perp E_N$, we have in particular

$$f - S_N f \perp S_N f.$$

Therefore, by the Pythagorean identity,

$$\|f\|_{L^2(\mathbb{T})}^2 = \|S_N f\|_{L^2(\mathbb{T})}^2 + \|f - S_N f\|_{L^2(\mathbb{T})}^2.$$

On the other hand,

$$\|S_N f\|_{L^2(\mathbb{T})}^2 = \left\langle \sum_{|k|\leqslant N} \widehat{f}(k) e_k, \sum_{|m|\leqslant N} \widehat{f}(m) e_m \right\rangle.$$

By linearity in the first variable and conjugate linearity in the second,

$$\|S_N f\|_{L^2(\mathbb{T})}^2 = \sum_{|k|\leqslant N} \sum_{|m|\leqslant N} \widehat{f}(k)\,\overline{\widehat{f}(m)}\,\langle e_k, e_m\rangle.$$

Since $\langle e_k, e_m\rangle = \delta_{k,m}$, all cross terms vanish, and we obtain

$$\|S_N f\|_{L^2(\mathbb{T})}^2 = \sum_{|k|\leqslant N} |\widehat{f}(k)|^2.$$

Since $S_N f \to f$ in $L^2(\mathbb{T})$, we have

$$\|f - S_N f\|_{L^2(\mathbb{T})} \longrightarrow 0.$$

Passing to the limit as $N \to \infty$ in

$$\|f\|_{L^2(\mathbb{T})}^2 = \|S_N f\|_{L^2(\mathbb{T})}^2 + \|f - S_N f\|_{L^2(\mathbb{T})}^2,$$

we obtain

$$\|f\|_{L^2(\mathbb{T})}^2 = \lim_{N\to\infty} \sum_{|k|\leqslant N} |\widehat{f}(k)|^2 = \sum_{k\in\mathbb{Z}} |\widehat{f}(k)|^2.$$

This is Parseval's identity. ■

Remark 1.14

In Definition 1.10, the symbol $\sim$ indicates that the Fourier series is, in general, a *formal expansion*. Its precise meaning is given by Theorem 1.13, which states that for every $f \in L^2(\mathbb{T})$,

$$S_N f \longrightarrow f \qquad \text{in } L^2(\mathbb{T}),$$

that is,

$$\|f - S_N f\|_{L^2(\mathbb{T})} \longrightarrow 0.$$

This means that the mean square error between f and its partial Fourier sums tends to zero.

However, this convergence is *not* pointwise in general. In particular, it does not imply that

$$S_N f(\theta) \longrightarrow f(\theta) \quad \text{for every } \theta \in [-\pi, \pi].$$

Indeed, functions in $L^2(\mathbb{T})$ are defined only up to sets of measure zero, and the Fourier series may converge to a representative of the equivalence class different from the original pointwise function.

Moreover, without additional regularity assumptions on f, the Fourier series may fail to converge at certain points. Stronger forms of convergence (pointwise or uniform) require further hypotheses on f (for instance continuity or piecewise regularity).

Example 2 : Euler identity via Parseval

We illustrate Parseval's identity by recovering the classical formula

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Consider the function $f(\theta) = \theta$ on $[-\pi, \pi]$, extended as a 2π-periodic function. Then $f \in L^2(\mathbb{T})$ and is odd, so its Fourier series formally reduces to a sine series :

$$f(\theta) \sim \sum_{n=1}^{\infty} b_n \sin(n\theta).$$

The Fourier coefficients are given by

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \theta \sin(n\theta)\, d\theta.$$

Since the integrand is even, we obtain

$$b_n = \frac{2}{\pi} \int_0^{\pi} \theta \sin(n\theta)\, d\theta.$$

An integration by parts yields

$$\int_0^{\pi} \theta \sin(n\theta)\, d\theta = \left[-\frac{\theta \cos(n\theta)}{n} \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos(n\theta)\, d\theta.$$

The second term vanishes, hence

$$\int_0^{\pi} \theta \sin(n\theta)\, d\theta = -\frac{\pi \cos(n\pi)}{n} = -\frac{\pi(-1)^n}{n}.$$

Therefore,

$$b_n = \frac{2}{\pi} \times \left(-\frac{\pi(-1)^n}{n} \right) = \frac{2(-1)^{n+1}}{n}.$$

We now compute the $L^2(\mathbb{T})$ norm of f :

$$\|f\|^2_{L^2(\mathbb{T})} = \int_{-\pi}^{\pi} \theta^2 \, \frac{d\theta}{2\pi} = \frac{1}{2\pi} \times 2\int_0^{\pi} \theta^2 \, d\theta = \frac{1}{\pi} \times \frac{\pi^3}{3} = \frac{\pi^2}{3}.$$

On the other hand, using Parseval's identity (Theorem 1.13), we have

$$\|f\|^2_{L^2(\mathbb{T})} = \sum_{k\in\mathbb{Z}} |\widehat{f}(k)|^2.$$

Using Proposition 1.11, one obtains

$$\widehat{f}(n) = \frac{b_n}{2i}, \qquad \widehat{f}(-n) = -\frac{b_n}{2i}, \qquad \widehat{f}(0) = 0.$$

Therefore,

$$|\widehat{f}(n)|^2 = \frac{|b_n|^2}{4}, \qquad |\widehat{f}(-n)|^2 = \frac{|b_n|^2}{4},$$

and Parseval's identity yields

$$\|f\|^2_{L^2(\mathbb{T})} = \sum_{n=1}^{\infty} \Big(|\widehat{f}(n)|^2 + |\widehat{f}(-n)|^2\Big) = \frac{1}{2}\sum_{n=1}^{\infty} |b_n|^2.$$

Substituting $b_n = \dfrac{2(-1)^{n+1}}{n}$, we obtain

$$\frac{\pi^2}{3} = \frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{2}{n}\right)^2 = 2\sum_{n=1}^{\infty}\frac{1}{n^2}.$$

Hence,

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}.$$

6) Riemann–Lebesgue Lemma

This is the periodic analogue of the Riemann–Lebesgue lemma for the Fourier transform on $\mathbb{R}$.

Proposition 1.15: Riemann–Lebesgue Lemma

If $f \in L^1(\mathbb{T})$, then

$$\widehat{f}(k) \longrightarrow 0 \qquad \text{as } |k| \to \infty.$$

Proof:

We first prove the result for trigonometric polynomials. Let

$$P(\theta) = \sum_{|n|\leqslant N} c_n e^{in\theta}.$$

Then, by the orthogonality relations,

$$\widehat{P}(k) = \begin{cases} c_k, & |k| \leqslant N, \\ 0, & |k| > N. \end{cases}$$

Hence

$$\widehat{P}(k) \longrightarrow 0 \qquad \text{as } |k| \to \infty.$$

Now let $f \in L^1(\mathbb{T})$, and let $\varepsilon > 0$. Since trigonometric polynomials are dense in $L^1(\mathbb{T})$, there exists a trigonometric polynomial P such that

$$\|f - P\|_{L^1(\mathbb{T})} < \varepsilon.$$

For every $k \in \mathbb{Z}$, we have

$$|\widehat{f}(k) - \widehat{P}(k)| = \left| \int_{-\pi}^{\pi} (f(\theta) - P(\theta)) e^{-ik\theta} \frac{d\theta}{2\pi} \right|.$$

Since $|e^{-ik\theta}| = 1$, it follows that

$$|\widehat{f}(k) - \widehat{P}(k)| \leqslant \int_{-\pi}^{\pi} |f(\theta) - P(\theta)| \frac{d\theta}{2\pi} = \|f - P\|_{L^1(\mathbb{T})} < \varepsilon.$$

On the other hand, since $\widehat{P}(k) \to 0$, there exists $K > 0$ such that

$$|k| \geqslant K \quad \Longrightarrow \quad |\widehat{P}(k)| < \varepsilon.$$

Therefore, for $|k| \geqslant K$,

$$|\widehat{f}(k)| \leqslant |\widehat{f}(k) - \widehat{P}(k)| + |\widehat{P}(k)| < 2\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we conclude that

$$\widehat{f}(k) \longrightarrow 0 \qquad \text{as } |k| \to \infty.$$

■

7) Convolution on the Circle

Definition 1.16

If $f, g \in L^1(\mathbb{T})$, their convolution is defined by

$$(f * g)(\theta) = \int_{-\pi}^{\pi} f(\theta - t) g(t) \frac{dt}{2\pi}.$$

Proposition 1.17

If $f, g \in L^1(\mathbb{T})$, then $f * g$ belongs to $L^1(\mathbb{T})$. Moreover,

$$\|f * g\|_{L^1(\mathbb{T})} \leqslant \|f\|_{L^1(\mathbb{T})} \|g\|_{L^1(\mathbb{T})}.$$

Proof:
For $\theta \in \mathbb{R}$, we have

$$|(f * g)(\theta)| \leqslant \int_{-\pi}^{\pi} |f(\theta - t)| \, |g(t)| \frac{dt}{2\pi}.$$

Integrating over $\theta \in [-\pi, \pi]$ and using Tonelli's theorem,

$$\int_{-\pi}^{\pi} |f * g(\theta)| \frac{d\theta}{2\pi} \leqslant \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} |f(\theta - t)| \, |g(t)| \frac{dt}{2\pi} \frac{d\theta}{2\pi}.$$

Using the change of variable $u = \theta - t$ and periodicity,

$$\int_{-\pi}^{\pi} |f(\theta - t)| \frac{d\theta}{2\pi} = \int_{-\pi}^{\pi} |f(u)| \frac{du}{2\pi}.$$

Thus,

$$\|f * g\|_{L^1(\mathbb{T})} \leqslant \int_{-\pi}^{\pi} |g(t)| \frac{dt}{2\pi} \int_{-\pi}^{\pi} |f(u)| \frac{du}{2\pi} = \|f\|_{L^1} \|g\|_{L^1}.$$

■

Remark 1.18

Thus $(L^1(\mathbb{T}), *, \|\cdot\|_1)$ is a Banach algebra : it is stable under convolution, and it is complete with respect to the norm $\|\cdot\|_1$.

Proposition 1.19: Commutativity of convolution

Let $f, g \in L^1(\mathbb{T})$. Then
$$f * g = g * f.$$

Proof:
For $\theta \in \mathbb{R}$, we have
$$(f * g)(\theta) = \int_{-\pi}^{\pi} f(\theta - t)\, g(t)\, \frac{dt}{2\pi}.$$
We perform the change of variable
$$u = \theta - t, \qquad dt = -du.$$
When $t = -\pi$, we have $u = \theta + \pi$, and when $t = \pi$, we have $u = \theta - \pi$. Thus
$$(f * g)(\theta) = \int_{\theta+\pi}^{\theta-\pi} f(u)\, g(\theta - u)\, \frac{-du}{2\pi} = \int_{\theta-\pi}^{\theta+\pi} f(u)\, g(\theta - u)\, \frac{du}{2\pi}.$$
Since f and g are 2π-periodic, the integral over $[\theta - \pi, \theta + \pi]$ is equal to the integral over $[-\pi, \pi]$. Hence
$$(f * g)(\theta) = \int_{-\pi}^{\pi} f(u)\, g(\theta - u)\, \frac{du}{2\pi}.$$
Interchanging the roles of f and g, we obtain
$$(f * g)(\theta) = (g * f)(\theta).$$

■

Proposition 1.20

If $f, g \in L^1(\mathbb{T})$, then for every $k \in \mathbb{Z}$,
$$\widehat{f * g}(k) = \hat{f}(k)\hat{g}(k).$$

Proof:
By definition,
$$\widehat{f * g}(k) = \int_{-\pi}^{\pi} \left(\int_{-\pi}^{\pi} f(\theta - t) g(t)\, \frac{dt}{2\pi} \right) e^{-ik\theta}\, \frac{d\theta}{2\pi}.$$
By Fubini's theorem,
$$\widehat{f * g}(k) = \int_{-\pi}^{\pi} g(t) \left(\int_{-\pi}^{\pi} f(\theta - t) e^{-ik\theta}\, \frac{d\theta}{2\pi} \right) \frac{dt}{2\pi}.$$
Set $u = \theta - t$. Since f is 2π-periodic, this gives
$$\int_{-\pi}^{\pi} f(\theta - t) e^{-ik\theta}\, \frac{d\theta}{2\pi} = e^{-ikt} \int_{-\pi}^{\pi} f(u) e^{-iku}\, \frac{du}{2\pi} = e^{-ikt}\hat{f}(k).$$
Therefore
$$\widehat{f * g}(k) = \hat{f}(k) \int_{-\pi}^{\pi} g(t) e^{-ikt}\, \frac{dt}{2\pi} = \hat{f}(k)\hat{g}(k).$$

■

8) Remark and Interpretation

Remark 1.21

Although our Fourier series theory has been developed on the torus

$$\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z},$$

that is, for 2π-periodic functions, it extends naturally to functions of arbitrary period $T > 0$.

This extension can be described as follows :

— If F is T-periodic, define a 2π-periodic function

$$G(\theta) := F\left(\frac{T}{2\pi}\theta\right).$$

— The function G belongs to the framework of $\mathbb{T}$, hence admits a Fourier expansion

$$G(\theta) \sim \sum_{n\in\mathbb{Z}} \widehat{G}(n)e^{in\theta}.$$

— Returning to the original variable $x = \frac{T}{2\pi}\theta$, we obtain

$$F(x) \sim \sum_{n\in\mathbb{Z}} \widehat{F}(n)e^{2\pi inx/T},$$

with coefficients

$$\widehat{F}(n) = \frac{1}{T}\int_{-T/2}^{T/2} F(x)e^{-2\pi inx/T}\,dx \overset{\text{by periodicity}}{=} \frac{1}{T}\int_0^T F(x)e^{-2\pi inx/T}\,dx.$$

Thus, Fourier series on $\mathbb{R}/T\mathbb{Z}$ are obtained from those on $\mathbb{T}$ by a simple scaling argument.

Fourier series provide a decomposition of periodic functions into elementary oscillations :

$$f(\theta) \sim \sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ik\theta}.$$

This decomposition admits the following interpretation :

— Each function $e^{ik\theta}$ represents a pure oscillation of frequency k.

— The coefficient $\widehat{f}(k)$ encodes both the amplitude and the phase of this oscillation.

— The sequence $(\widehat{f}(k))_{k\in\mathbb{Z}}$ describes the *frequency content* of f.

Hence, a function f can be viewed as a superposition of elementary waves, a fundamental viewpoint in applications such as signal processing and physics.

This decomposition is discrete due to the compactness of the domain :

— The circle $\mathbb{T}$ is compact.

— As a consequence, only discrete frequencies $k \in \mathbb{Z}$ appear.

The Fourier transform on $\mathbb{R}$ will arise as a continuous analogue of this theory. Formally, one replaces :

$$k \in \mathbb{Z} \quad \longrightarrow \quad \xi \in \mathbb{R}, \qquad \sum_{k\in\mathbb{Z}} \quad \longrightarrow \quad \int_{-\infty}^{\infty}.$$

In this transition :

— The discrete spectrum $(\widehat{f}(k))$ becomes a continuous function $\widehat{f}(\xi)$.

— The Fourier series becomes the Fourier inversion formula

$$f(x) = \int_{-\infty}^{\infty} \widehat{f}(\xi)\, e^{2\pi i x\xi}\, d\xi.$$

A rigorous justification of this passage will be developed in the next sections.

II - From Fourier Series to Fourier Transform

The theory of Fourier series applies to functions defined on the circle, or equivalently, to periodic functions on $\mathbb{R}$. We now seek an analogous theory for functions defined on the whole real line, without any periodicity assumption.

For a periodic function, the Fourier expansion associates to f a sequence of coefficients $\left(\widehat{f}(k)\right)_{k\in\mathbb{Z}}$, which describe the contribution of each discrete frequency. For a function defined on $\mathbb{R}$, it is therefore natural to expect an analogous representation involving a *continuous* range of frequencies, leading to a function $\widehat{f}$ defined on $\mathbb{R}$.

In the 1-periodic setting, one writes

$$\widehat{f}(k) = \int_0^1 f(x)e^{-2\pi i k x}\, dx, \qquad k \in \mathbb{Z},$$

and, at least formally,

$$f(x) \sim \sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{2\pi i k x}.$$

Replacing the discrete variable k by a continuous variable $\xi \in \mathbb{R}$, and the sum by an integral, naturally suggests the definition of the Fourier transform :

$$\widehat{f}(\xi) = \int_{-\infty}^{\infty} f(t)e^{-2\pi i \xi t}\, dt, \qquad \xi \in \mathbb{R},$$

together with the formal inversion formula

$$f(x) \sim \int_{-\infty}^{\infty} \widehat{f}(\xi)e^{2\pi i x\xi}\, d\xi.$$

Thus, while Fourier series associate to a function a discrete sequence of coefficients indexed by $\mathbb{Z}$, the Fourier transform associates to a function on $\mathbb{R}$ another function on $\mathbb{R}$:

$$f \longleftrightarrow \widehat{f}.$$

This transition from a discrete to a continuous frequency representation is one of the central features of Fourier analysis, and will be made rigorous in the following sections.

III - First Insight

We now explain how the Fourier inversion formula can be formally recovered from Fourier series through the process of periodization.

Definition 3.1

Let $f : \mathbb{R} \to \mathbb{C}$ be a compactly supported function and let $T > 0$. The T-periodization of f is the function

$$F(x) := \sum_{k\in\mathbb{Z}} f(x + kT).$$

Proposition 3.2

If $f \in \mathcal{C}_c^1(\mathbb{R})$ (the space of $\mathcal{C}^1$ functions with compact support), then for every $T > 0$, the above sum contains only finitely many nonzero terms for each x, and the function F is T-periodic and of class $\mathcal{C}^1$.

Proof:
Since f has compact support, there exists $R > 0$ such that $f(y) = 0$ whenever $|y| > R$. For fixed x, the condition $f(x + kT) \neq 0$ implies $|x + kT| \leqslant R$, hence only finitely many integers k can occur. Thus the sum defining $F(x)$ is finite.
Periodicity follows from

$$F(x + T) = \sum_{k\in\mathbb{Z}} f(x + T + kT) = \sum_{k\in\mathbb{Z}} f(x + kT) = F(x).$$

Since locally the sum is finite and each summand is of class $\mathcal{C}^1$, the function F is of class $\mathcal{C}^1$. ■
Applying Fourier series theory to F on the interval $[0, T]$, one writes

$$F(x) = \sum_{n\in\mathbb{Z}} \widehat{F}(n)e^{2\pi inx/T},$$

where the renormalized Fourier coefficients are

$$\widehat{F}(n) = \int_0^T F(x)e^{-2\pi inx/T}\,\frac{dx}{T}.$$

Proposition 3.3

Let $f \in \mathcal{C}_c^1(\mathbb{R})$, let F be its T-periodization, and define

$$\widehat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i\xi x}\,dx.$$

Then, for every $n \in \mathbb{Z}$,

$$\widehat{F}(n) = \frac{1}{T}\widehat{f}\left(\frac{n}{T}\right).$$

Proof:
By definition,

$$\widehat{F}(n) = \int_0^T \left(\sum_{k\in\mathbb{Z}} f(x + kT)\right) e^{-2\pi inx/T}\,\frac{dx}{T}.$$

Since only finitely many terms are nonzero for each x, we may interchange sum and integral :

$$\widehat{F}(n) = \frac{1}{T}\sum_{k\in\mathbb{Z}} \int_0^T f(x+kT)e^{-2\pi inx/T}\,dx.$$

Set $u = x + kT$. Then $dx = du$, and

$$e^{-2\pi inx/T} = e^{-2\pi in(u-kT)/T} = e^{-2\pi inu/T}e^{2\pi ink} = e^{-2\pi inu/T}.$$

Therefore

$$\widehat{F}(n) = \frac{1}{T}\sum_{k\in\mathbb{Z}} \int_{kT}^{(k+1)T} f(u)e^{-2\pi inu/T}\,du.$$

Summing over all k yields

$$\widehat{F}(n) = \frac{1}{T}\int_{-\infty}^{\infty} f(u)e^{-2\pi i(n/T)u}\,du = \frac{1}{T}\widehat{f}\left(\frac{n}{T}\right).$$

■ If, moreover, $\operatorname{supp}(f) \subset [0, T]$, then $F(x) = f(x)$ on $[0, T]$, so the Fourier series of F reads

$$f(x) = \frac{1}{T}\sum_{n\in\mathbb{Z}} \widehat{f}\left(\frac{n}{T}\right)e^{2\pi inx/T}.$$

This is a Riemann sum, and letting $T \to \infty$ suggests the inversion formula

$$f(x) = \int_{-\infty}^{\infty} \widehat{f}(\xi)e^{2\pi ix\xi}\,d\xi.$$

Before formulating a rigorous version, we recall the natural space involved.

Definition 3.4

The space $\mathcal{C}_c^1(\mathbb{R})$ consists of all functions $f : \mathbb{R} \to \mathbb{C}$ such that :

1. $f \in \mathcal{C}^1(\mathbb{R})$;
2. f has compact support, that is,

$$\operatorname{supp}(f) := \overline{\{x \in \mathbb{R} : f(x) \neq 0\}}$$

is a compact subset of $\mathbb{R}$.

Theorem 3.5

Let $f \in \mathcal{C}_c^1(\mathbb{R})$. Its Fourier transform

$$\widehat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i\xi x}\,dx$$

is continuous as a function of ξ with values in $\mathbb{C}$.
Moreover, if, in addition, $\widehat{f}$ is Riemann integrable on $\mathbb{R}$, then the original function f is recovered by the Fourier inversion formula :

$$f(x) = \int_{-\infty}^{\infty} \widehat{f}(\xi)e^{2\pi ix\xi}\,d\xi.$$

Proof:

Step 1 : Continuity of $\widehat{f}$.

Let $\xi_0 \in \mathbb{R}$ and let (ξ_n) be a sequence such that $\xi_n \to \xi_0$. For each $x \in \mathbb{R}$,

$$f(x)e^{-2\pi i\xi_n x} \longrightarrow f(x)e^{-2\pi i\xi_0 x}.$$

Moreover, since $|e^{-2\pi i\xi x}| = 1$, we have

$$|f(x)e^{-2\pi i\xi_n x}| = |f(x)|.$$

Since $f \in \mathcal{C}_c^1(\mathbb{R}) \subset L^1(\mathbb{R})$, the function $|f(x)|$ is integrable. Therefore, by the dominated convergence theorem,

$$\widehat{f}(\xi_n) = \int_{\mathbb{R}} f(x)e^{-2\pi i\xi_n x}\,dx \longrightarrow \int_{\mathbb{R}} f(x)e^{-2\pi i\xi_0 x}\,dx = \widehat{f}(\xi_0).$$

Hence $\widehat{f}$ is continuous.

Step 2 : Periodization and Fourier series.

Let F be the T-periodization of f (Definition 3.1) :

$$F(x) = \sum_{k\in\mathbb{Z}} f(x+kT).$$

Since $f \in \mathcal{C}_c^1(\mathbb{R})$, Proposition 3.2 ensures that the sum contains only finitely many nonzero terms for each x, and that F is T-periodic and of class $\mathcal{C}^1$.

Since F is T-periodic and of class $\mathcal{C}^1$, its Fourier series converges pointwise to F, and thus

$$F(x) = \sum_{n\in\mathbb{Z}} \widehat{F}(n)e^{2\pi inx/T},$$

where

$$\widehat{F}(n) = \frac{1}{T}\int_0^T F(t)e^{-2\pi int/T}\,dt.$$

By the computation of the Fourier coefficients of the periodization (Proposition 3.3), we have

$$\widehat{F}(n) = \frac{1}{T}\widehat{f}\left(\frac{n}{T}\right).$$

Now fix $x \in \mathbb{R}$. Since $\operatorname{supp}(f)$ is compact, one can choose $c \in \mathbb{R}$ and $T > 0$ such that

$$\operatorname{supp}(f) \subset (c, c+T) \qquad \text{and} \qquad x \in [c, c+T].$$

For this choice of T, the periodization F satisfies

$$F(x) = f(x).$$

Therefore, evaluating the Fourier series of F at this point x, we obtain

$$f(x) = \frac{1}{T}\sum_{n\in\mathbb{Z}} \widehat{f}\left(\frac{n}{T}\right) e^{2\pi inx/T}.$$

Step 3 : Passage to the limit.

Set $\xi_n = \frac{n}{T}$ for $n \in \mathbb{Z}$. Then the identity obtained above can be rewritten as

$$f(x) = \sum_{n\in\mathbb{Z}} \widehat{f}(\xi_n)e^{2\pi ix\xi_n}\,\frac{1}{T}.$$

Observe that the points $(\xi_n)_{n\in\mathbb{Z}}$ form a uniform partition of $\mathbb{R}$ with step size

$$\xi_{n+1} - \xi_n = \frac{1}{T}.$$

Hence, the real line is discretized into intervals of equal length $\frac{1}{T}$, and the above sum can be written in the form

$$\sum_{n\in\mathbb{Z}} g(\xi_n)\,\Delta\xi, \qquad \text{with } \Delta\xi = \frac{1}{T},$$

where

$$g(\xi) = \widehat{f}(\xi)e^{2\pi i x\xi}.$$

This is precisely a Riemann sum associated with the function g on $\mathbb{R}$. Indeed, it approximates the integral of g over $\mathbb{R}$ by sampling the function at the grid points ξ_n and multiplying by the mesh size $\frac{1}{T}$.
As $T \to \infty$, the mesh size $\frac{1}{T}$ tends to 0, and the grid (ξ_n) becomes dense in $\mathbb{R}$. In this limit, the Riemann sums formally converge to the integral of g, namely

$$\int_{-\infty}^{\infty} \widehat{f}(\xi)e^{2\pi i x\xi}\,d\xi.$$

Under the additional assumption that $\widehat{f}$ is Riemann integrable on $\mathbb{R}$, this leads to the Fourier inversion formula

$$f(x) = \int_{-\infty}^{\infty} \widehat{f}(\xi)e^{2\pi i x\xi}\,d\xi.$$

■

IV - Fourier Transform of Functions with Moderate Growth

We now extend the definition of the Fourier transform beyond compactly supported functions. The first natural class consists of functions that decay sufficiently fast at infinity to ensure convergence of integrals.

1) Improper Integrals on $\mathbb{R}$

Definition 4.1

Let $f : \mathbb{R} \to \mathbb{C}$ be continuous. We define the improper integral of f over $\mathbb{R}$ by

$$\int_{-\infty}^{\infty} f(x)\,dx := \lim_{M\to\infty} \int_{-M}^{0} f(x)\,dx + \lim_{N\to\infty} \int_{0}^{N} f(x)\,dx,$$

whenever both limits exist and are finite.

This definition extends the Riemann integral to unbounded intervals.

2) Functions of Moderate Growth

Definition 4.2

A continuous function $f : \mathbb{R} \to \mathbb{C}$ is said to be of *moderate growth* if there exists a constant $A > 0$ such that

$$|f(x)| \leqslant \frac{A}{1+x^2}, \qquad \forall x \in \mathbb{R}.$$

More generally, one may consider functions satisfying

$$|f(x)| \leqslant \frac{C}{1+|x|^{1+\varepsilon}}, \qquad \varepsilon > 0,$$

but we restrict ourselves to the case $\varepsilon = 1$ for simplicity.

Proposition 4.3

Every function of moderate growth is integrable on $\mathbb{R}$.

Proof:
Let f satisfy

$$|f(x)| \leqslant \frac{A}{1+x^2}.$$

Then

$$\int_{-\infty}^{\infty} |f(x)|\,dx \leqslant A \int_{-\infty}^{\infty} \frac{dx}{1+x^2}.$$

Since

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi < \infty,$$

the result follows. ∎

3) Definition of the Fourier Transform

Definition 4.4

Let $f : \mathbb{R} \to \mathbb{C}$ be a continuous function of moderate growth. Its Fourier transform is defined by

$$\widehat{f}(\xi) := \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x}\, dx, \qquad \xi \in \mathbb{R}.$$

We denote the Fourier transform of f by $\widehat{f}$, or equivalently by $\mathcal{F}(f)$.

Proposition 4.5

The Fourier transform $\widehat{f}$ is bounded and satisfies

$$|\widehat{f}(\xi)| \leqslant \int_{-\infty}^{\infty} |f(x)|\, dx.$$

Proof:
Since $|e^{-2\pi i \xi x}| = 1$, we have

$$|\widehat{f}(\xi)| = \left| \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x}\, dx \right| \leqslant \int_{-\infty}^{\infty} |f(x)|\, dx.$$

■

Proposition 4.6

If f is continuous and of moderate growth, then $\widehat{f}$ is continuous on $\mathbb{R}$.

Proof:
Let $\xi_n \to \xi$. Then

$$f(x) e^{-2\pi i \xi_n x} \to f(x) e^{-2\pi i \xi x} \quad \text{pointwise.}$$

Moreover,

$$|f(x) e^{-2\pi i \xi_n x}| \leqslant |f(x)|,$$

and $f \in L^1(\mathbb{R})$. By the dominated convergence theorem,

$$\widehat{f}(\xi_n) \to \widehat{f}(\xi).$$

■

4) Riemann–Lebesgue Property

Theorem 4.7

If f is continuous and of moderate growth, then

$$\widehat{f}(\xi) \longrightarrow 0 \quad \text{as } |\xi| \to \infty.$$

This result extends the Riemann–Lebesgue lemma from Fourier series to the Fourier transform.

However, this class of functions is not stable under differentiation, which motivates the introduction of a more suitable functional space.

V - Fourier Transform in the Schwartz Space $\mathcal{S}(\mathbb{R})$

The previous definition of the Fourier transform for functions with moderate growth is only partially satisfactory, since the defining integral may fail to converge in general.
To obtain a fully rigorous and stable theory, we now introduce the Schwartz space $\mathcal{S}(\mathbb{R})$, which provides an ideal framework where the Fourier transform is well-defined, infinitely differentiable, and enjoys strong decay properties.
This space will serve as the fundamental setting from which the Fourier transform will later be extended to $L^1(\mathbb{R})$ and $L^2(\mathbb{R})$.

1) Definition of the Schwartz Space

Definition 5.1

The Schwartz space $\mathcal{S}(\mathbb{R})$ is the set of all functions $f \in \mathcal{C}^\infty(\mathbb{R})$ such that for all integers $k, \ell \geqslant 0$,

$$\sup_{x\in\mathbb{R}} |x^k f^{(\ell)}(x)| < \infty.$$

Functions in $\mathcal{S}(\mathbb{R})$ are infinitely differentiable and rapidly decreasing at infinity, together with all their derivatives.

Remark 5.2

The quantities

$$p_{k,\ell}(f) = \sup_{x\in\mathbb{R}} |x^k f^{(\ell)}(x)|, \qquad k, \ell \in \mathbb{N},$$

are called *seminorms* on $\mathcal{S}(\mathbb{R})$. They satisfy positivity, homogeneity, and the triangle inequality, but they are not norms since, for a fixed pair (k, ℓ), the condition $p_{k,\ell}(f) = 0$ does not necessarily imply that $f = 0$.
This family of seminorms defines the *Schwartz topology*.

Proposition 5.3

$\mathcal{S}(\mathbb{R})$ is a vector space stable under differentiation and multiplication by polynomials :

$$f \in \mathcal{S}(\mathbb{R}) \Rightarrow f' \in \mathcal{S}(\mathbb{R}), \quad xf(x) \in \mathcal{S}(\mathbb{R}).$$

Proof:
Recall that $f \in \mathcal{S}(\mathbb{R})$ means that $f \in \mathcal{C}^\infty(\mathbb{R})$ and that, for every pair of integers $m, n \geqslant 0$,

$$\sup_{x\in\mathbb{R}} |x^m f^{(n)}(x)| < \infty.$$

We first show that $\mathcal{S}(\mathbb{R})$ is a vector space. Let $f, g \in \mathcal{S}(\mathbb{R})$ and let $\alpha, \beta \in \mathbb{C}$. Then $\alpha f + \beta g \in \mathcal{C}^\infty(\mathbb{R})$, and for every $m, n \geqslant 0$,

$$x^m(\alpha f + \beta g)^{(n)}(x) = \alpha x^m f^{(n)}(x) + \beta x^m g^{(n)}(x).$$

Hence

$$|x^m(\alpha f + \beta g)^{(n)}(x)| \leqslant |\alpha|\,|x^m f^{(n)}(x)| + |\beta|\,|x^m g^{(n)}(x)|.$$

Since the two quantities on the right are bounded on $\mathbb{R}$, it follows that

$$\sup_{x\in\mathbb{R}} |x^m(\alpha f + \beta g)^{(n)}(x)| < \infty.$$

Therefore $\alpha f + \beta g \in \mathcal{S}(\mathbb{R})$.

Next, let $f \in \mathcal{S}(\mathbb{R})$. We prove that $f' \in \mathcal{S}(\mathbb{R})$. Since $f \in \mathcal{C}^\infty(\mathbb{R})$, we clearly have $f' \in \mathcal{C}^\infty(\mathbb{R})$. Now let $m, n \geqslant 0$. Then

$$(f')^{(n)} = f^{(n+1)},$$

so

$$x^m (f')^{(n)}(x) = x^m f^{(n+1)}(x).$$

Because $f \in \mathcal{S}(\mathbb{R})$, the right-hand side is bounded on $\mathbb{R}$. Hence

$$\sup_{x\in\mathbb{R}} |x^m (f')^{(n)}(x)| < \infty.$$

Thus $f' \in \mathcal{S}(\mathbb{R})$.

Finally, we prove that $xf(x) \in \mathcal{S}(\mathbb{R})$. Since $f \in \mathcal{C}^\infty(\mathbb{R})$, the function $x \mapsto xf(x)$ is also of class $\mathcal{C}^\infty(\mathbb{R})$. Let $m, n \geqslant 0$. By Leibniz's formula, since the derivatives of the function $x \mapsto x$ vanish after order 1, we have

$$(xf)^{(n)}(x) = xf^{(n)}(x) + nf^{(n-1)}(x) \qquad (n \geqslant 1),$$

and for $n = 0$,

$$(xf)^{(0)}(x) = xf(x).$$

Therefore, for $n \geqslant 1$,

$$x^m (xf)^{(n)}(x) = x^{m+1} f^{(n)}(x) + nx^m f^{(n-1)}(x).$$

Since $f \in \mathcal{S}(\mathbb{R})$, both terms on the right-hand side are bounded on $\mathbb{R}$. Hence

$$\sup_{x\in\mathbb{R}} |x^m (xf)^{(n)}(x)| < \infty.$$

For $n = 0$, we simply have

$$x^m (xf)(x) = x^{m+1} f(x),$$

which is also bounded because $f \in \mathcal{S}(\mathbb{R})$.
Thus $xf \in \mathcal{S}(\mathbb{R})$, and the proof is complete. ■

2) Integrability Properties

Lemma 5.4

If $f \in \mathcal{S}(\mathbb{R})$, then for all integers $k, \ell \geqslant 0$,

$$\int_{-\infty}^{\infty} |x^k f^{(\ell)}(x)|\, dx < \infty.$$

Proof:
By definition,

$$|x^k f^{(\ell)}(x)| \leqslant \frac{M}{1+x^2},$$

for some constant M, hence integrability follows from comparison with $\frac{1}{1+x^2}$. ■

3) Definition of the Fourier Transform

Definition 5.5

For $f \in \mathcal{S}(\mathbb{R})$, define
$$\widehat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x}\, dx.$$

4) Elementary Properties

The following proposition collects the fundamental properties of the Fourier transform, which will be used throughout the sequel.

Proposition 5.6

Let $f, g \in \mathcal{S}(\mathbb{R})$ and $\lambda, \mu \in \mathbb{C}$. Then :

i **Linearity :**
$$\mathcal{F}(\lambda f + \mu g)(\xi) = \lambda\, \widehat{f}(\xi) + \mu\, \widehat{g}(\xi).$$

ii **Translation :**
$$\mathcal{F}\big(f(\cdot + h)\big)(\xi) = e^{2\pi i \xi h}\widehat{f}(\xi).$$

iii **Modulation :**
$$\mathcal{F}\big(e^{-2\pi i h x} f(x)\big)(\xi) = \widehat{f}(\xi + h).$$

iv **Dilation :**
$$\mathcal{F}\big(f(\delta x)\big)(\xi) = \frac{1}{|\delta|}\widehat{f}\left(\frac{\xi}{\delta}\right), \qquad \delta \neq 0.$$

v **Differentiation :**
$$\mathcal{F}(f')(\xi) = 2\pi i \xi\, \widehat{f}(\xi).$$

More generally, for every integer $k \geqslant 0$,
$$\widehat{f^{(k)}}(\xi) = (2\pi i \xi)^k \widehat{f}(\xi).$$

vi **Multiplication by x :**
$$\mathcal{F}(xf)(\xi) = \frac{i}{2\pi}\frac{d}{d\xi}\widehat{f}(\xi).$$

More generally, for every integer $\ell \geqslant 0$,
$$\mathcal{F}\big(x^\ell f\big)(\xi) = \left(\frac{i}{2\pi}\right)^\ell \frac{d^\ell}{d\xi^\ell}\widehat{f}(\xi).$$

vii **Convolution :**
$$\widehat{f * g}(\xi) = \widehat{f}(\xi)\, \widehat{g}(\xi).$$

Moreover, for every $g \in L^1(\mathbb{R})$, we have
$$\|\widehat{g}\|_{L^\infty(\mathbb{R})} \leqslant \|g\|_{L^1(\mathbb{R})}.$$

Proof:

We prove each property. **(i) Linearity.**

$$\begin{aligned}\mathcal{F}(\lambda f+\mu g)(\xi) &= \int_{\mathbb{R}} (\lambda f(x)+\mu g(x))e^{-2\pi i x\xi}\,dx \\ &= \lambda \widehat{f}(\xi)+\mu\widehat{g}(\xi).\end{aligned}$$

(ii) Translation.

$$\mathcal{F}(f(\cdot+h))(\xi) = \int_{\mathbb{R}} f(x+h)e^{-2\pi i x\xi}\,dx.$$

With $y=x+h$, we get

$$\mathcal{F}(f(\cdot+h))(\xi) = \int_{\mathbb{R}} f(y)e^{-2\pi i(y-h)\xi}\,dy = e^{2\pi i\xi h}\widehat{f}(\xi).$$

(iii) Modulation.

$$\mathcal{F}(e^{-2\pi i h x}f(x))(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi i x(\xi+h)}\,dx = \widehat{f}(\xi+h).$$

(iv) Dilation.

$$\mathcal{F}(f(\delta x))(\xi) = \int_{\mathbb{R}} f(\delta x)e^{-2\pi i x\xi}\,dx.$$

With $y=\delta x$, we have $dx = |\delta|^{-1}dy$, hence

$$\mathcal{F}(f(\delta x))(\xi) = \frac{1}{|\delta|}\int_{\mathbb{R}} f(y)e^{-2\pi i(y/\delta)\xi}\,dy = \frac{1}{|\delta|}\widehat{f}\left(\frac{\xi}{\delta}\right).$$

(v) Differentiation.

$$\widehat{f'}(\xi) = \int_{\mathbb{R}} f'(x)e^{-2\pi i x\xi}\,dx.$$

Integrating by parts and using $f(x)\to 0$ at infinity,

$$\widehat{f'}(\xi) = 2\pi i\xi\widehat{f}(\xi).$$

The general case follows by induction.

(vi) Multiplication by x.

$$\begin{aligned}\frac{d}{d\xi}\widehat{f}(\xi) &= \int_{\mathbb{R}} f(x)(-2\pi i x)e^{-2\pi i x\xi}\,dx \\ &= -2\pi i\,\widehat{xf}(\xi).\end{aligned}$$

Hence

$$\widehat{xf}(\xi) = \frac{i}{2\pi}\frac{d}{d\xi}\widehat{f}(\xi).$$

The general formula follows by induction.

(vii) Convolution. This property is the continuous analogue of the convolution formula on the torus (Proposition 1.20). Since $f,g\in\mathcal{S}(\mathbb{R})$, we have $f,g\in L^1(\mathbb{R})$, and Fubini's theorem applies. Therefore,

$$\begin{aligned}\widehat{f*g}(\xi) &= \int_{\mathbb{R}} (f*g)(x)e^{-2\pi i x\xi}\,dx \\ &= \int_{\mathbb{R}}\left(\int_{\mathbb{R}} f(x-y)g(y)\,dy\right)e^{-2\pi i x\xi}\,dx.\end{aligned}$$

By Fubini's theorem,

$$\widehat{f * g}(\xi) = \int_{\mathbb{R}} g(y) \left(\int_{\mathbb{R}} f(x-y) e^{-2\pi i x \xi} \, dx \right) dy.$$

Set $u = x - y$. Then $x = u + y$ and $dx = du$. Hence

$$\int_{\mathbb{R}} f(x-y) e^{-2\pi i x \xi} \, dx = \int_{\mathbb{R}} f(u) e^{-2\pi i (u+y) \xi} \, du = e^{-2\pi i y \xi} \widehat{f}(\xi).$$

$$\begin{aligned} \widehat{f * g}(\xi) &= \widehat{f}(\xi) \int_{\mathbb{R}} g(y) e^{-2\pi i y \xi} \, dy \\ &= \widehat{f}(\xi) \widehat{g}(\xi). \end{aligned}$$

Finally, if $g \in L^1(\mathbb{R})$,

$$|\widehat{g}(\xi)| \leqslant \int_{\mathbb{R}} |g(x)| dx = \|g\|_{L^1(\mathbb{R})}.$$

Taking the supremum gives the result. ■

5) Stability of the Schwartz Space

Theorem 5.7

If $f \in \mathcal{S}(\mathbb{R})$, then

$$\widehat{f} \in \mathcal{S}(\mathbb{R}).$$

Proof:
Let $k, \ell \in \mathbb{N}$. We want to show that

$$\sup_{\xi \in \mathbb{R}} \left| \xi^k \frac{d^\ell}{d\xi^\ell} \widehat{f}(\xi) \right| < \infty.$$

First, by the differentiation property in Proposition 5.6, we have

$$\frac{d^\ell}{d\xi^\ell} \widehat{f}(\xi) = \mathcal{F}\big((-2\pi i x)^\ell f(x) \big)(\xi).$$

Next, using the identity

$$\widehat{g^{(k)}}(\xi) = (2\pi i \xi)^k \widehat{g}(\xi),$$

we obtain

$$\xi^k \widehat{g}(\xi) = \frac{1}{(2\pi i)^k} \widehat{g^{(k)}}(\xi).$$

Applying this with

$$g(x) = (-2\pi i x)^\ell f(x),$$

we get

$$\begin{aligned} \xi^k \frac{d^\ell}{d\xi^\ell} \widehat{f}(\xi) &= \xi^k \mathcal{F}\big(g\big)(\xi) \\ &= \frac{1}{(2\pi i)^k} \widehat{g^{(k)}}(\xi) \\ &= \frac{1}{(2\pi i)^k} \mathcal{F}\left(\frac{d^k}{dx^k} \big[(-2\pi i x)^\ell f(x) \big] \right)(\xi). \end{aligned}$$

Define

$$H_{k,\ell}(x) = \frac{1}{(2\pi i)^k} \frac{d^k}{dx^k}\big[(-2\pi i x)^\ell f(x)\big].$$

Then

$$\xi^k \frac{d^\ell}{d\xi^\ell} \widehat{f}(\xi) = \widehat{H_{k,\ell}}(\xi).$$

By Leibniz's formula, $H_{k,\ell}$ is a finite linear combination of terms of the form

$$x^a f^{(b)}(x),$$

with $a, b \in \mathbb{N}$. Since $f \in \mathcal{S}(\mathbb{R})$, each such function is rapidly decreasing and therefore belongs to $L^1(\mathbb{R})$. Hence

$$H_{k,\ell} \in L^1(\mathbb{R}).$$

Applying the estimate

$$\|\widehat{g}\|_\infty \leqslant \|g\|_{L^1},$$

we obtain

$$\sup_{\xi \in \mathbb{R}} \left| \xi^k \frac{d^\ell}{d\xi^\ell} \widehat{f}(\xi) \right| = \|\widehat{H_{k,\ell}}\|_\infty \leqslant \|H_{k,\ell}\|_{L^1} < \infty.$$

Since this holds for all $k, \ell \in \mathbb{N}$, all the seminorms of $\widehat{f}$ are finite. Hence

$$\widehat{f} \in \mathcal{S}(\mathbb{R}).$$

■

6) The Gaussian Function

Proposition 5.8

For every $a > 0$, the function

$$x \longmapsto e^{-ax^2}$$

belongs to $\mathcal{S}(\mathbb{R})$.

Proof:
Let

$$f_a(x) := e^{-ax^2}, \qquad a > 0.$$

By induction on $\ell \geqslant 0$, one shows that there exists a polynomial P_ℓ of degree at most ℓ such that

$$\frac{d^\ell}{dx^\ell}\big(e^{-ax^2}\big) = P_\ell(x) e^{-ax^2}.$$

Thus, for every $k, \ell \geqslant 0$,

$$x^k f_a^{(\ell)}(x) = x^k P_\ell(x) e^{-ax^2}.$$

Since the exponential factor e^{-ax^2} decays faster than any power of $|x|$ as $|x| \to \infty$, the function $x^k f_a^{(\ell)}(x)$ is bounded on $\mathbb{R}$. Hence

$$\sup_{x \in \mathbb{R}} |x^k f_a^{(\ell)}(x)| < \infty,$$

which proves that $f_a \in \mathcal{S}(\mathbb{R})$. ■
Among all Gaussian functions, the normalized Gaussian

$$G(x) := e^{-\pi x^2}$$

plays a distinguished role.

Proposition 5.9

One has
$$\int_{-\infty}^{\infty} e^{-\pi x^2}\,dx = 1.$$

Proof:
Set
$$I := \int_{-\infty}^{\infty} e^{-\pi x^2}\,dx.$$
Then
$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\pi(x^2+y^2)}\,dx\,dy.$$
Passing to polar coordinates,
$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-\pi r^2} r\,dr\,d\theta = 2\pi\int_0^{\infty} e^{-\pi r^2} r\,dr.$$
Now
$$2\pi\int_0^{\infty} e^{-\pi r^2} r\,dr = \left[-e^{-\pi r^2}\right]_0^{\infty} = 1.$$
Therefore $I^2 = 1$. Since $I > 0$, we conclude that $I = 1$. ■

7) The Fourier Transform of the Gaussian

Theorem 5.10

Let
$$G(x) := e^{-\pi x^2}.$$
Then
$$\widehat{G}(\xi) = G(\xi) = e^{-\pi \xi^2}.$$

Proof:
Set
$$F(\xi) := \widehat{G}(\xi) = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\,dx.$$
By the previous proposition,
$$F(0) = \int_{-\infty}^{\infty} e^{-\pi x^2}\,dx = 1.$$
Since $G \in \mathcal{S}(\mathbb{R})$, we may differentiate under the integral sign :
$$F'(\xi) = \int_{-\infty}^{\infty} \underbrace{e^{-\pi x^2}(-2\pi i x)}_{=iG'(x)} e^{-2\pi i x\xi}\,dx.$$
Hence
$$F'(\xi) = i\int_{-\infty}^{\infty} G'(x) e^{-2\pi i x\xi}\,dx = i\widehat{G'}(\xi).$$
Using the differentiation formula for the Fourier transform,
$$\widehat{G'}(\xi) = 2\pi i\xi\,\widehat{G}(\xi) = 2\pi i\xi\,F(\xi),$$

we obtain

$$F'(\xi) = -2\pi\xi F(\xi).$$

Thus F satisfies the differential equation

$$F'(\xi) + 2\pi\xi F(\xi) = 0.$$

Its solutions are of the form

$$F(\xi) = Ce^{-\pi\xi^2}.$$

Since $F(0) = 1$, we get $C = 1$, and therefore

$$F(\xi) = e^{-\pi\xi^2}.$$

This proves that $\widehat{G} = G$. ■

8) Scaled Gaussian Kernels

Definition 5.11

For $\delta > 0$, define

$$K_\delta(x) := \frac{1}{\sqrt{\delta}}\, e^{-\frac{\pi x^2}{\delta}}.$$

Proposition 5.12

For every $\delta > 0$,

$$\widehat{K_\delta}(\xi) = e^{-\pi\delta\xi^2}.$$

Equivalently,

$$\widehat{K_\delta}(\xi) = \frac{1}{\sqrt{\delta}}\, K_{1/\delta}(\xi).$$

Proof:
Observe that

$$K_\delta(x) = \frac{1}{\sqrt{\delta}}\, G\left(\frac{x}{\sqrt{\delta}}\right).$$

Using the dilation property of the Fourier transform,

$$f(\lambda x) \longmapsto \frac{1}{|\lambda|}\hat{f}\left(\frac{\xi}{\lambda}\right),$$

with $\lambda = \frac{1}{\sqrt{\delta}}$, and the fact that $\widehat{G} = G$, we get

$$\widehat{K_\delta}(\xi) = e^{-\pi\delta\xi^2}.$$

Since

$$K_{1/\delta}(\xi) = \sqrt{\delta}\, e^{-\pi\delta\xi^2},$$

it follows that

$$\widehat{K_\delta}(\xi) = \frac{1}{\sqrt{\delta}}\, K_{1/\delta}(\xi).$$

■

9) Gaussian Approximation of the Identity

Lemma 5.13: Properties of the Gaussian kernels

The family $(K_\delta)_{\delta>0}$ satisfies :

i $K_\delta(x) > 0$ for all $x \in \mathbb{R}$;

ii

$$\int_{-\infty}^{\infty} K_\delta(x)\,dx = 1;$$

iii for every fixed $\eta > 0$,

$$\lim_{\delta\to 0}\int_{|x|\geqslant\eta} K_\delta(x)\,dx = 0.$$

Proof:

Property (i) is immediate from the definition.
For (ii), we use the change of variables $x = \sqrt{\delta}\,y$:

$$\int_{-\infty}^{\infty} K_\delta(x)\,dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{\delta}} e^{-\frac{\pi x^2}{\delta}}\,dx = \int_{-\infty}^{\infty} e^{-\pi y^2}\,dy = 1.$$

For (iii), let $\eta > 0$. Again with $x = \sqrt{\delta}\,y$,

$$\int_{|x|\geqslant\eta} K_\delta(x)\,dx = 2\int_{\eta}^{\infty} \frac{1}{\sqrt{\delta}} e^{-\frac{\pi x^2}{\delta}}\,dx = 2\int_{\eta/\sqrt{\delta}}^{\infty} e^{-\pi y^2}\,dy.$$

Set

$$A = \frac{\eta}{\sqrt{\delta}}.$$

As $\delta \to 0$, we have $A \to \infty$. We claim that

$$\int_{A}^{\infty} e^{-\pi y^2}\,dy \longrightarrow 0 \qquad \text{as } A \to \infty.$$

Indeed, for $y \geqslant A > 0$, we have $y^2 \geqslant Ay$, hence

$$e^{-\pi y^2} \leqslant e^{-\pi A y}.$$

Therefore,

$$\int_{A}^{\infty} e^{-\pi y^2}\,dy \leqslant \int_{A}^{\infty} e^{-\pi A y}\,dy = \frac{e^{-\pi A^2}}{\pi A} \longrightarrow 0.$$

Consequently,

$$\int_{\eta/\sqrt{\delta}}^{\infty} e^{-\pi y^2}\,dy \longrightarrow 0,$$

and thus

$$\int_{|x|\geqslant\eta} K_\delta(x)\,dx \longrightarrow 0.$$

■

Definition 5.14

For $f \in L^1(\mathbb{R})$ and $\delta > 0$, define the Gaussian regularization of f by

$$(f * K_\delta)(x) = \int_{-\infty}^{\infty} f(x-y)K_\delta(y)\,dy.$$

Proposition 5.15

If $f \in \mathcal{S}(\mathbb{R})$, then for every $x \in \mathbb{R}$,

$$(f * K_\delta)(x) \longrightarrow f(x) \qquad \text{as } \delta \to 0.$$

Proof:
We write

$$(f * K_\delta)(x) - f(x) = \int_{-\infty}^{\infty} \big(f(x-y) - f(x)\big) K_\delta(y)\, dy.$$

Fix $\varepsilon > 0$. Since f is continuous, there exists $\eta > 0$ such that

$$|y| < \eta \quad \Longrightarrow \quad |f(x-y) - f(x)| < \varepsilon.$$

Hence

$$|(f * K_\delta)(x) - f(x)| \leqslant \int_{|y|<\eta} |f(x-y) - f(x)| K_\delta(y)\, dy + \int_{|y|\geqslant\eta} |f(x-y) - f(x)| K_\delta(y)\, dy.$$

For the first term,

$$\int_{|y|<\eta} |f(x-y) - f(x)| K_\delta(y)\, dy \leqslant \varepsilon \int_{|y|<\eta} K_\delta(y)\, dy \leqslant \varepsilon.$$

For the second term, since f is bounded,

$$|f(x-y) - f(x)| \leqslant 2\|f\|_\infty,$$

so

$$\int_{|y|\geqslant\eta} |f(x-y) - f(x)| K_\delta(y)\, dy \leqslant 2\|f\|_\infty \int_{|y|\geqslant\eta} K_\delta(y)\, dy.$$

By the previous lemma, this tends to 0 as $\delta \to 0$. Therefore

$$\limsup_{\delta\to 0} |(f * K_\delta)(x) - f(x)| \leqslant \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, the result follows. ■

10) Fourier Inversion in the Schwartz Space

Theorem 5.16: Fourier inversion formula

If $f \in \mathcal{S}(\mathbb{R})$, then for every $x \in \mathbb{R}$,

$$f(x) = \int_{-\infty}^{\infty} \widehat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

Proof:
Since $f \in \mathcal{S}(\mathbb{R})$, we have $\widehat{f} \in \mathcal{S}(\mathbb{R}) \subset L^1(\mathbb{R})$. Hence the integral

$$\int_{-\infty}^{\infty} \widehat{f}(\xi) e^{2\pi i x\xi}\, d\xi$$

is well-defined.

For $\delta > 0$, define

$$I_\delta(x) := \int_{-\infty}^{\infty} \widehat{f}(\xi) e^{-\pi\delta\xi^2} e^{2\pi i x\xi}\, d\xi.$$

Using the definition of the Fourier transform,

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(y) e^{-2\pi i y\xi}\, dy,$$

we obtain

$$I_\delta(x) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(y) e^{-2\pi i y\xi}\, dy \right) e^{-\pi\delta\xi^2} e^{2\pi i x\xi}\, d\xi.$$

By Fubini's theorem,

$$I_\delta(x) = \int_{\mathbb{R}} f(y) \left(\int_{\mathbb{R}} e^{-\pi\delta\xi^2} e^{2\pi i(x-y)\xi}\, d\xi \right) dy.$$

By the Gaussian Fourier transform formula established in Proposition 5.12, with $t = x - y$, we have

$$\int_{\mathbb{R}} e^{-\pi\delta\xi^2} e^{2\pi i t\xi}\, d\xi = \widehat{e^{-\pi\delta\xi^2}}(-t) = \frac{1}{\sqrt{\delta}} e^{-\pi t^2/\delta} = K_\delta(t).$$

Therefore,

$$I_\delta(x) = \int_{\mathbb{R}} f(y) K_\delta(x-y)\, dy = (f * K_\delta)(x).$$

By the properties of the approximate identity (K_δ),

$$I_\delta(x) \longrightarrow f(x) \qquad \text{as } \delta \to 0.$$

On the other hand, for every ξ,

$$\widehat{f}(\xi) e^{-\pi\delta\xi^2} e^{2\pi i x\xi} \longrightarrow \widehat{f}(\xi) e^{2\pi i x\xi},$$

and

$$\left| \widehat{f}(\xi) e^{-\pi\delta\xi^2} e^{2\pi i x\xi} \right| \leqslant |\widehat{f}(\xi)|.$$

Since $\widehat{f} \in L^1(\mathbb{R})$, the dominated convergence theorem yields

$$I_\delta(x) \longrightarrow \int_{-\infty}^{\infty} \widehat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

Comparing the two limits, we conclude that

$$f(x) = \int_{-\infty}^{\infty} \widehat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

■

11) The Inverse Fourier Transform

Definition 5.17

For $g \in \mathcal{S}(\mathbb{R})$, define the inverse Fourier transform by

$$\check{g}(x) := \int_{-\infty}^{\infty} g(\xi) e^{2\pi i x\xi}\, d\xi.$$

Proposition 5.18

Let $f \in \mathcal{S}(\mathbb{R})$. Then :

i

$$\check{\hat{f}} = f.$$

ii

$$\check{f}(x) = \hat{f}(-x), \qquad x \in \mathbb{R}.$$

iii The Fourier transform is injective on $\mathcal{S}(\mathbb{R})$. More precisely, if

$$f, g \in \mathcal{S}(\mathbb{R}) \qquad \text{and} \qquad \hat{f} = \hat{g},$$

then

$$f = g.$$

iv

$$\hat{\check{f}} = f.$$

Proof:
(i) The identity

$$\check{\hat{f}} = f$$

is precisely the Fourier inversion formula.

(ii) By definition of the inverse Fourier transform,

$$\check{f}(x) = \int_{-\infty}^{\infty} f(\xi) e^{2\pi i x\xi}\, d\xi.$$

On the other hand,

$$\hat{f}(-x) = \int_{-\infty}^{\infty} f(\xi) e^{-2\pi i\xi(-x)}\, d\xi = \int_{-\infty}^{\infty} f(\xi) e^{2\pi i x\xi}\, d\xi.$$

Hence

$$\check{f}(x) = \hat{f}(-x).$$

(iii) Assume that

$$\hat{f} = \hat{g}.$$

Applying the inverse Fourier transform to both sides yields

$$\check{\hat{f}} = \check{\hat{g}}.$$

Using part **(i)**, we obtain

$$f = g.$$

In particular, if $\hat{f} = 0$, then

$$f = \check{\hat{f}} = 0.$$

Hence the Fourier transform is injective on $\mathcal{S}(\mathbb{R})$.

(iv) Let $f \in \mathcal{S}(\mathbb{R})$. Applying part **(i)** to $\check{f}$, we obtain

$$\check{\check{f}} = \check{f}.$$

Since the inverse Fourier transform is injective by part **(iii)**, it follows that

$$\hat{\check{f}} = f.$$

■

Lemma 5.19

The Fourier transform maps the Schwartz space onto itself :

$$\mathcal{F}\big(\mathcal{S}(\mathbb{R})\big) = \mathcal{S}(\mathbb{R}).$$

Proof:
First, by Theorem 5.7, we already know that

$$f \in \mathcal{S}(\mathbb{R}) \quad \Longrightarrow \quad \hat{f} \in \mathcal{S}(\mathbb{R}).$$

Hence

$$\mathcal{F}\big(\mathcal{S}(\mathbb{R})\big) \subset \mathcal{S}(\mathbb{R}).$$

Conversely, let $g \in \mathcal{S}(\mathbb{R})$. Since the inverse Fourier transform also maps $\mathcal{S}(\mathbb{R})$ into itself, we have

$$f := \check{g} \in \mathcal{S}(\mathbb{R}).$$

By the identity proved above,

$$\hat{\check{g}} = g.$$

Thus there exists $f \in \mathcal{S}(\mathbb{R})$, namely $f = \check{g}$, such that

$$\hat{f} = g.$$

Therefore,

$$\mathcal{S}(\mathbb{R}) \subset \mathcal{F}\big(\mathcal{S}(\mathbb{R})\big).$$

Combining the two inclusions gives

$$\mathcal{F}\big(\mathcal{S}(\mathbb{R})\big) = \mathcal{S}(\mathbb{R}).$$

■

VI - The Fourier Transform on $L^1(\mathbb{R})$

Before developing the Fourier transform in the Hilbert space setting $L^2(\mathbb{R})$, it is natural to first define it for integrable functions. This approach follows the historical development of the theory and provides a pointwise definition with strong analytical properties.

Recall that

$$L^1(\mathbb{R}) = \left\{ f : \mathbb{R} \to \mathbb{C} \text{ measurable} : \int_{-\infty}^{\infty} |f(x)|\, dx < \infty \right\}.$$

Definition 6.1: Fourier transform on $L^1(\mathbb{R})$

Let $f \in L^1(\mathbb{R})$. The Fourier transform of f is defined by

$$\hat{f}(\xi) := \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\, dx, \qquad \xi \in \mathbb{R}.$$

This integral is well-defined since

$$|f(x)e^{-2\pi i x\xi}| = |f(x)|,$$

and $f \in L^1(\mathbb{R})$.

Let us now establish the fundamental properties of this transform.

Proposition 6.2: Basic properties of the L^1 Fourier transform

Let $f \in L^1(\mathbb{R})$. Then :

i $\widehat{f}$ is bounded :

$$|\widehat{f}(\xi)| \leqslant \int_{-\infty}^{\infty} |f(x)|\, dx = \|f\|_{L^1(\mathbb{R})}.$$

ii $\widehat{f}$ is uniformly continuous on $\mathbb{R}$.

iii The Fourier transform is linear : for all $a, b \in \mathbb{C}$,

$$\widehat{(af + bg)} = a\widehat{f} + b\widehat{g}.$$

Proof:
(i) follows immediately from the triangle inequality :

$$|\widehat{f}(\xi)| \leqslant \int |f(x)|\, dx.$$

(ii) Let $h \to 0$. Then

$$\widehat{f}(\xi + h) - \widehat{f}(\xi) = \int f(x) \left(e^{-2\pi i x(\xi+h)} - e^{-2\pi i x\xi}\right) dx.$$

The integrand tends pointwise to 0, and is dominated by

$$2|f(x)| \in L^1(\mathbb{R}).$$

By the dominated convergence theorem,

$$\widehat{f}(\xi + h) \to \widehat{f}(\xi).$$

(iii) follows from linearity of the integral. ■

Remark 6.3

The translation, modulation, and dilation properties established on $\mathcal{S}(\mathbb{R})$ (Proposition 5.6) remain valid for the Fourier transform on $L^1(\mathbb{R})$.

The differentiation and multiplication formulas require additional regularity and integrability assumptions.

VII - The Fourier Transform on $L^2(\mathbb{R})$

We have defined the Fourier transform on the Schwartz space $\mathcal{S}(\mathbb{R})$, proved that it maps $\mathcal{S}(\mathbb{R})$ into itself, and established the inversion formula there. We now show that this transform extends naturally to the Hilbert space $L^2(\mathbb{R})$.

1) A First Identity on the Schwartz Space

Theorem 7.1: Plancherel identity on $\mathcal{S}(\mathbb{R})$

For every $f, g \in \mathcal{S}(\mathbb{R})$,

$$\langle \widehat{f}, \widehat{g} \rangle_{L^2(\mathbb{R})} = \langle f, g \rangle_{L^2(\mathbb{R})}.$$

In particular,

$$\|\widehat{f}\|_{L^2(\mathbb{R})} = \|f\|_{L^2(\mathbb{R})}.$$

Thus, on $\mathcal{S}(\mathbb{R})$, the Fourier transform preserves the L^2 inner product.
Proof:
Since $f, g \in \mathcal{S}(\mathbb{R})$, all functions involved belong to $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, and we may apply the inversion formula to g :

$$g(x) = \int_{-\infty}^{\infty} \widehat{g}(\xi) e^{2\pi i x \xi} \, d\xi.$$

Taking complex conjugates, we obtain

$$\overline{g(x)} = \int_{-\infty}^{\infty} \overline{\widehat{g}(\xi)} \, e^{-2\pi i x \xi} \, d\xi.$$

Hence

$$\int_{-\infty}^{\infty} f(x)\overline{g(x)} \, dx = \int_{-\infty}^{\infty} f(x) \left(\int_{-\infty}^{\infty} \overline{\widehat{g}(\xi)} \, e^{-2\pi i x \xi} \, d\xi \right) dx.$$

By Fubini's theorem,

$$\int_{-\infty}^{\infty} f(x)\overline{g(x)} \, dx = \int_{-\infty}^{\infty} \overline{\widehat{g}(\xi)} \left(\int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} \, dx \right) d\xi.$$

The inner integral is exactly $\widehat{f}(\xi)$, hence

$$\int_{-\infty}^{\infty} f(x)\overline{g(x)} \, dx = \int_{-\infty}^{\infty} \widehat{f}(\xi)\overline{\widehat{g}(\xi)} \, d\xi.$$

Taking $g = f$, we obtain

$$\int_{-\infty}^{\infty} |\widehat{f}(\xi)|^2 \, d\xi = \int_{-\infty}^{\infty} |f(x)|^2 \, dx,$$

that is,

$$\|\widehat{f}\|_{L^2(\mathbb{R})} = \|f\|_{L^2(\mathbb{R})}.$$

■

2) Density of the Schwartz Space in $L^2(\mathbb{R})$

Proposition 7.2

The Schwartz space $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$.

We shall admit this classical result here.

This density property is fundamental, since it allows us to extend the Fourier transform from the Schwartz space to the whole Hilbert space $L^2(\mathbb{R})$.

3) Extension to $L^2(\mathbb{R})$

The Parseval identity established on $\mathcal{S}(\mathbb{R})$

$$\langle \widehat{f}, \widehat{g} \rangle_{L^2(\mathbb{R})} = \langle f, g \rangle_{L^2(\mathbb{R})}$$

shows that the Fourier transform preserves the L^2 inner product and, in particular, the L^2 norm on the Schwartz space :

$$\|\widehat{f}\|_{L^2(\mathbb{R})} = \|f\|_{L^2(\mathbb{R})}.$$

This identity is the continuous analogue of Parseval's formula for Fourier series on the torus :

$$\|f\|^2_{L^2(\mathbb{T})} = \sum_{k \in \mathbb{Z}} |\widehat{f}(k)|^2.$$

Since $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, the Fourier transform extends uniquely by continuity to the whole space $L^2(\mathbb{R})$.

Theorem 7.3: Plancherel theorem

There exists a unique linear operator

$$\mathcal{F} : L^2(\mathbb{R}) \longrightarrow L^2(\mathbb{R})$$

extending the Fourier transform on $\mathcal{S}(\mathbb{R})$ and satisfying

$$\langle \mathcal{F}(f), \mathcal{F}(g) \rangle_{L^2(\mathbb{R})} = \langle f, g \rangle_{L^2(\mathbb{R})}, \qquad f, g \in L^2(\mathbb{R}).$$

In particular,

$$\|\mathcal{F}(f)\|_{L^2(\mathbb{R})} = \|f\|_{L^2(\mathbb{R})}, \qquad f \in L^2(\mathbb{R}).$$

Proof:

Let $f \in L^2(\mathbb{R})$. Since $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, there exists a sequence $(f_n)_{n \geqslant 1} \subset \mathcal{S}(\mathbb{R})$ such that

$$f_n \to f \qquad \text{in } L^2(\mathbb{R}).$$

By the Parseval identity on $\mathcal{S}(\mathbb{R})$,

$$\|\widehat{f_n} - \widehat{f_m}\|_{L^2} = \|f_n - f_m\|_{L^2}.$$

Hence $(\widehat{f_n})_{n \geqslant 1}$ is a Cauchy sequence in the complete space $L^2(\mathbb{R})$. Therefore there exists $g \in L^2(\mathbb{R})$ such that

$$\widehat{f_n} \to g \qquad \text{in } L^2(\mathbb{R}).$$

We define

$$\mathcal{F}(f) := \lim_{n \to \infty} \widehat{f_n} \qquad \text{in } L^2(\mathbb{R}).$$

If $(h_n)_{n\geqslant 1} \subset \mathcal{S}(\mathbb{R})$ also satisfies
$$h_n \to f \qquad \text{in } L^2(\mathbb{R}),$$
then
$$\|\widehat{f_n} - \widehat{h_n}\|_{L^2} = \|f_n - h_n\|_{L^2} \leqslant \|f_n - f\|_{L^2} + \|h_n - f\|_{L^2} \to 0.$$
Hence the definition of $\mathcal{F}(f)$ is independent of the approximating sequence.

Let $f, g \in L^2(\mathbb{R})$ and $\lambda, \mu \in \mathbb{C}$. Choose sequences $(f_n)_{n\geqslant 1}$ and $(g_n)_{n\geqslant 1}$ in $\mathcal{S}(\mathbb{R})$ such that
$$f_n \to f, \qquad g_n \to g \qquad \text{in } L^2(\mathbb{R}).$$
Then
$$\lambda f_n + \mu g_n \to \lambda f + \mu g \qquad \text{in } L^2(\mathbb{R}).$$
By definition of the extension operator,
$$\mathcal{F}(f) = \lim_{n\to\infty} \widehat{f}_n, \qquad \mathcal{F}(g) = \lim_{n\to\infty} \widehat{g}_n.$$
Since the Fourier transform is linear on $\mathcal{S}(\mathbb{R})$,
$$\lambda \widehat{f_n + \mu g_n} = \lambda \widehat{f}_n + \mu \widehat{g}_n.$$
Using the continuity of addition and scalar multiplication in $L^2(\mathbb{R})$, we may pass to the limit and obtain
$$\mathcal{F}(\lambda f + \mu g) = \lambda \mathcal{F}(f) + \mu \mathcal{F}(g).$$
Hence $\mathcal{F}$ is linear.

Moreover,
$$\|\mathcal{F}(f)\|_{L^2} = \lim_{n\to\infty} \|\widehat{f}_n\|_{L^2} = \lim_{n\to\infty} \|f_n\|_{L^2} = \|f\|_{L^2}.$$
Finally, if $f, g \in L^2(\mathbb{R})$ and
$$f_n \to f, \qquad g_n \to g \qquad \text{in } L^2(\mathbb{R}),$$
with $f_n, g_n \in \mathcal{S}(\mathbb{R})$, then
$$\langle \mathcal{F}(f), \mathcal{F}(g) \rangle = \lim_{n\to\infty} \langle \widehat{f}_n, \widehat{g}_n \rangle.$$
Using the Parseval identity on $\mathcal{S}(\mathbb{R})$,
$$\langle \widehat{f}_n, \widehat{g}_n \rangle = \langle f_n, g_n \rangle.$$
Passing to the limit yields
$$\langle \mathcal{F}(f), \mathcal{F}(g) \rangle = \langle f, g \rangle.$$
This completes the proof. ■

4) Inversion on $L^2(\mathbb{R})$

Recall that, on the Schwartz space,
$$\check{\widehat{f}} = f, \qquad \widehat{\check{g}} = g, \qquad f, g \in \mathcal{S}(\mathbb{R}).$$
We now extend these identities to the whole space $L^2(\mathbb{R})$.

Theorem 7.4

The operator
$$\mathcal{F} : L^2(\mathbb{R}) \to L^2(\mathbb{R})$$
is unitary. Equivalently, it preserves the inner product :
$$\langle \mathcal{F}(f), \mathcal{F}(g) \rangle_{L^2(\mathbb{R})} = \langle f, g \rangle_{L^2(\mathbb{R})}, \qquad f, g \in L^2(\mathbb{R}),$$

and

$$\mathcal{F}(L^2(\mathbb{R})) = L^2(\mathbb{R}).$$

Its inverse is the continuous extension of the inverse Fourier transform from $\mathcal{S}(\mathbb{R})$ to $L^2(\mathbb{R})$.

Proof:
By the Plancherel theorem,

$$\langle \mathcal{F}(f), \mathcal{F}(g) \rangle = \langle f, g \rangle \qquad \forall f, g \in L^2(\mathbb{R}).$$

Hence $\mathcal{F}$ is an isometry. In particular, it is injective and its range is closed.

Moreover,

$$\mathcal{F}(\mathcal{S}(\mathbb{R})) = \mathcal{S}(\mathbb{R}),$$

and $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$. Therefore the range of $\mathcal{F}$ contains a dense subset of $L^2(\mathbb{R})$.
Since the range is both dense and closed, it follows that

$$\mathcal{F}(L^2(\mathbb{R})) = L^2(\mathbb{R}).$$

Thus $\mathcal{F}$ is onto (surjective).

Therefore $\mathcal{F}$ is a unitary operator on $L^2(\mathbb{R})$. Its inverse is the continuous extension of the inverse Fourier transform on $\mathcal{S}(\mathbb{R})$. ■

Remark 7.5

The Fourier transform is therefore a unitary automorphism of the Hilbert space $L^2(\mathbb{R})$. This viewpoint plays a fundamental role in harmonic analysis, partial differential equations, and quantum mechanics.

5) Summary

We have established the Fourier transform at three levels :

1. On $\mathcal{S}(\mathbb{R})$, it is defined by an absolutely convergent integral, preserves the Schwartz space, and satisfies the inversion formula.
2. On $L^1(\mathbb{R})$, it defines a bounded continuous function vanishing at infinity.
3. On $L^2(\mathbb{R})$, it extends uniquely to a unitary operator (Plancherel theorem).

To keep in mind

☛

$$C_c^\infty(\mathbb{R}) \subset \mathcal{S}(\mathbb{R}) \subset L^p(\mathbb{R}), \qquad 1 \leqslant p \leqslant \infty.$$

☛ $C_c^\infty(\mathbb{R})$ is dense in $\mathcal{S}(\mathbb{R})$, and $\mathcal{S}(\mathbb{R})$ is dense in $L^p(\mathbb{R})$ for $1 \leqslant p < \infty$.

☛ The general strategy is :

$$\text{prove on } \mathcal{S}(\mathbb{R}) \quad \Longrightarrow \quad \text{extend by density.}$$

☛ In particular, this yields the unitary extension of the Fourier transform to $L^2(\mathbb{R})$.

VIII - Tempered Distributions and the Fourier Transform

The Schwartz space $\mathcal{S}(\mathbb{R})$ is very well adapted to the Fourier transform : it is stable under differentiation, multiplication by polynomials, and Fourier transformation. Its topological dual provides the natural framework for extending the Fourier transform to generalized functions.

1) Test Functions and Seminorms

Definition 8.1

For $f \in \mathcal{S}(\mathbb{R})$ and integers $k, \ell \geqslant 0$, define the seminorm

$$p_{k,\ell}(f) := \sup_{x \in \mathbb{R}} (1 + |x|)^k |f^{(\ell)}(x)|.$$

These seminorms measure both the smoothness and the decay of a Schwartz function.

Definition 8.2

A sequence $(f_n)_{n \geqslant 1}$ in $\mathcal{S}(\mathbb{R})$ is said to converge to 0 in $\mathcal{S}(\mathbb{R})$ if

$$p_{k,\ell}(f_n) \longrightarrow 0 \qquad \text{for all } k, \ell \geqslant 0.$$

More generally, $f_n \to f$ in $\mathcal{S}(\mathbb{R})$ if

$$f_n - f \to 0 \qquad \text{in } \mathcal{S}(\mathbb{R}).$$

Thus, convergence in $\mathcal{S}(\mathbb{R})$ means uniform convergence of all derivatives, together with rapid decay at infinity.

2) Distributions on the Schwartz Space

Definition 8.3

A *tempered distribution* on $\mathbb{R}$ is a functional

$$T : \mathcal{S}(\mathbb{R}) \longrightarrow \mathbb{C}$$

satisfying the following properties :

1. **Linearity :** for every $\varphi, \psi \in \mathcal{S}(\mathbb{R})$ and every $\lambda, \mu \in \mathbb{C}$,

$$T(\lambda\varphi + \mu\psi) = \lambda T(\varphi) + \mu T(\psi).$$

2. **Continuity :** T is continuous with respect to the topology of $\mathcal{S}(\mathbb{R})$.

The space of all tempered distributions is denoted by

$$\mathcal{S}'(\mathbb{R}).$$

Proposition 8.4

A linear functional

$$T : \mathcal{S}(\mathbb{R}) \to \mathbb{C}$$

is continuous if and only if there exist integers $N, M \geqslant 0$ and a constant $C > 0$ such that

$$|T(\varphi)| \leqslant C \sum_{\substack{0\leqslant k\leqslant N \\ 0\leqslant \ell\leqslant M}} p_{k,\ell}(\varphi), \qquad \forall \varphi \in \mathcal{S}(\mathbb{R}).$$

We shall use this characterization without proof in the sequel.

3) Examples of Tempered Distributions

Example 3 : Every function $f \in \mathcal{S}(\mathbb{R})$ defines a tempered distribution T_f by

$$T_f(\varphi) = \int_{-\infty}^{\infty} f(x)\varphi(x)\,dx, \qquad \varphi \in \mathcal{S}(\mathbb{R}).$$

Proposition 8.5

If $f \in \mathcal{S}(\mathbb{R})$, then

$$T_f \in \mathcal{S}'(\mathbb{R}).$$

Proof:
Linearity is immediate.

Since $f \in \mathcal{S}(\mathbb{R})$, there exists a constant $C > 0$ such that

$$|f(x)| \leqslant \frac{C}{(1+|x|)^2}, \qquad x \in \mathbb{R}.$$

Let $\varphi \in \mathcal{S}(\mathbb{R})$. Then

$$|T_f(\varphi)| \leqslant \int_{\mathbb{R}} |f(x)\varphi(x)|\,dx \leqslant C \int_{\mathbb{R}} \frac{|\varphi(x)|}{(1+|x|)^2}\,dx.$$

Using

$$|\varphi(x)| \leqslant p_{2,0}(\varphi)\frac{1}{(1+|x|)^2},$$

we obtain

$$|T_f(\varphi)| \leqslant C\, p_{2,0}(\varphi) \int_{\mathbb{R}} \frac{dx}{(1+|x|)^4}.$$

Hence

$$|T_f(\varphi)| \leqslant C' p_{2,0}(\varphi),$$

for some constant $C' > 0$.
Therefore T_f is continuous on $\mathcal{S}(\mathbb{R})$, and consequently

$$T_f \in \mathcal{S}'(\mathbb{R}).$$

■

Example 4 : Every function

$$f \in L^1_{\mathrm{loc}}(\mathbb{R})$$

with at most polynomial growth defines a tempered distribution by

$$T_f(\varphi) = \int_{-\infty}^{\infty} f(x)\varphi(x)\,dx, \qquad \varphi \in \mathcal{S}(\mathbb{R}),$$

provided this integral is absolutely convergent.

Proposition 8.6

Let
$$f \in L^1_{\text{loc}}(\mathbb{R})$$
satisfy a polynomial growth condition, i.e. there exist constants $C > 0$ and $N \in \mathbb{N}$ such that
$$|f(x)| \leqslant C(1+|x|)^N, \qquad x \in \mathbb{R}.$$
Then the functional
$$T_f : \mathcal{S}(\mathbb{R}) \longrightarrow \mathbb{C},$$
defined by
$$T_f(\varphi) = \int_{-\infty}^{\infty} f(x)\varphi(x)\,dx, \qquad \varphi \in \mathcal{S}(\mathbb{R}),$$
is a tempered distribution.

Proof:
Linearity is immediate.

Let $\varphi \in \mathcal{S}(\mathbb{R})$. Since φ is rapidly decreasing, for every integer $m \geqslant 0$,
$$\sup_{x\in\mathbb{R}} (1+|x|)^m |\varphi(x)| < \infty.$$
In particular, choosing $m = N+2$, we obtain
$$|\varphi(x)| \leqslant \frac{p_{N+2,0}(\varphi)}{(1+|x|)^{N+2}}.$$
Using the growth condition on f,
$$|f(x)\varphi(x)| \leqslant C\, p_{N+2,0}(\varphi) \frac{1}{(1+|x|)^2}.$$
Since
$$\frac{1}{(1+|x|)^2} \in L^1(\mathbb{R}),$$
the integral defining $T_f(\varphi)$ is absolutely convergent.

Moreover,
$$|T_f(\varphi)| \leqslant C\, p_{N+2,0}(\varphi) \int_{\mathbb{R}} \frac{dx}{(1+|x|)^2}.$$
Hence
$$|T_f(\varphi)| \leqslant C' p_{N+2,0}(\varphi),$$
for some constant $C' > 0$.
Thus T_f is continuous on $\mathcal{S}(\mathbb{R})$, and therefore
$$T_f \in \mathcal{S}'(\mathbb{R}).$$
■

Example 5 : The Dirac mass at the origin is the functional
$$\delta_0(\varphi) := \varphi(0).$$
More generally, for every $a \in \mathbb{R}$,
$$\delta_a(\varphi) := \varphi(a).$$

Proposition 8.7

For every $a \in \mathbb{R}$, the Dirac distribution δ_a is a tempered distribution.

Proof:
Linearity follows immediately from the definition.

Let $\varphi \in \mathcal{S}(\mathbb{R})$. From the definition of δ_a,

$$\delta_a(\varphi) = \varphi(a).$$

Hence

$$|\delta_a(\varphi)| = |\varphi(a)| \leqslant \sup_{x\in\mathbb{R}} |\varphi(x)| = p_{0,0}(\varphi).$$

Therefore δ_a is continuous on $\mathcal{S}(\mathbb{R})$, and consequently

$$\delta_a \in \mathcal{S}'(\mathbb{R}).$$

■

Remark 8.8

The Dirac distribution is not represented by an ordinary function. This illustrates that tempered distributions extend the notion of classical functions and allow singular objects to be treated within the same framework.

4) Differentiation of Tempered Distributions

Definition 8.9

Let $T \in \mathcal{S}'(\mathbb{R})$. The derivative T' of T is the tempered distribution defined by

$$T'(\varphi) := -T(\varphi'), \qquad \varphi \in \mathcal{S}(\mathbb{R}).$$

More generally, for $m \in \mathbb{N}$,

$$T^{(m)}(\varphi) := (-1)^m T(\varphi^{(m)}).$$

Proposition 8.10

If $T \in \mathcal{S}'(\mathbb{R})$, then $T' \in \mathcal{S}'(\mathbb{R})$.

Proof:
The map

$$\varphi \longmapsto \varphi'$$

is a continuous linear map from $\mathcal{S}(\mathbb{R})$ into itself. Since T is continuous and linear, so is

$$\varphi \longmapsto -T(\varphi').$$

Hence $T' \in \mathcal{S}'(\mathbb{R})$. ■

Example 6 : The derivative of the Dirac mass is defined by

$$\delta_0'(\varphi) := -\varphi'(0).$$

More generally,

$$\delta_0^{(m)}(\varphi) = (-1)^m \varphi^{(m)}(0).$$

Proposition 8.11

If $f \in \mathcal{S}(\mathbb{R})$, then the distributional derivative of T_f agrees with the classical derivative :

$$(T_f)' = T_{f'}.$$

Proof:
For every $\varphi \in \mathcal{S}(\mathbb{R})$,

$$(T_f)'(\varphi) = -T_f(\varphi') = -\int_{-\infty}^{\infty} f(x)\varphi'(x)\,dx.$$

Since $f, \varphi \in \mathcal{S}(\mathbb{R})$, the boundary terms vanish at infinity, and integration by parts gives

$$-\int_{-\infty}^{\infty} f(x)\varphi'(x)\,dx = \int_{-\infty}^{\infty} f'(x)\varphi(x)\,dx = T_{f'}(\varphi).$$

Therefore

$$(T_f)' = T_{f'}.$$

■

5) Multiplication by the Variable

Definition 8.12

Let $T \in \mathcal{S}'(\mathbb{R})$. The tempered distribution xT is defined by

$$(xT)(\varphi) := T(x\varphi), \qquad \varphi \in \mathcal{S}(\mathbb{R}).$$

More generally, for $m \in \mathbb{N}$,

$$(x^m T)(\varphi) := T(x^m \varphi).$$

Proposition 8.13

If $T \in \mathcal{S}'(\mathbb{R})$, then $xT \in \mathcal{S}'(\mathbb{R})$.

Proof:
Since the map

$$\varphi \longmapsto x\varphi$$

is a continuous linear map from $\mathcal{S}(\mathbb{R})$ into itself, the functional

$$\varphi \longmapsto T(x\varphi)$$

is continuous and linear. Hence $xT \in \mathcal{S}'(\mathbb{R})$. ■

6) The Fourier Transform of a Tempered Distribution

Definition 8.14

Let $T \in \mathcal{S}'(\mathbb{R})$. The Fourier transform of T, denoted by $\widehat{T}$, is defined by

$$\widehat{T}(\varphi) := T(\widehat{\varphi}), \qquad \varphi \in \mathcal{S}(\mathbb{R}).$$

This definition is natural : since the Fourier transform maps $\mathcal{S}(\mathbb{R})$ onto itself, one may compose any tempered distribution with the Fourier transform of test functions.

Proposition 8.15

If $T \in \mathcal{S}'(\mathbb{R})$, then $\widehat{T} \in \mathcal{S}'(\mathbb{R})$.

Proof:
The Fourier transform

$$\varphi \longmapsto \widehat{\varphi}$$

is a continuous linear map from $\mathcal{S}(\mathbb{R})$ into itself. Since T is continuous and linear on $\mathcal{S}(\mathbb{R})$, the map

$$\varphi \longmapsto T(\widehat{\varphi})$$

is again a continuous linear functional on $\mathcal{S}(\mathbb{R})$. Hence $\widehat{T} \in \mathcal{S}'(\mathbb{R})$. ■

7) Compatibility with Classical Fourier Transform

Proposition 8.16

If $f \in \mathcal{S}(\mathbb{R})$, then the Fourier transform of the tempered distribution T_f is the distribution associated with the classical Fourier transform :

$$\widehat{T_f} = T_{\widehat{f}}.$$

Proof:
Let $\varphi \in \mathcal{S}(\mathbb{R})$. Then

$$\widehat{T_f}(\varphi) = T_f(\widehat{\varphi}) = \int_{-\infty}^{\infty} f(x)\widehat{\varphi}(x)\, dx.$$

Now

$$\widehat{\varphi}(x) = \int_{-\infty}^{\infty} \varphi(\xi)e^{-2\pi i x\xi}\, d\xi,$$

so by Fubini's theorem,

$$\widehat{T_f}(\varphi) = \int_{-\infty}^{\infty} \varphi(\xi)\left(\int_{-\infty}^{\infty} f(x)e^{-2\pi i x\xi}\, dx\right) d\xi.$$

The inner integral is $\widehat{f}(\xi)$, hence

$$\widehat{T_f}(\varphi) = \int_{-\infty}^{\infty} \widehat{f}(\xi)\varphi(\xi)\, d\xi = T_{\widehat{f}}(\varphi).$$

Therefore

$$\widehat{T_f} = T_{\widehat{f}}.$$

■

8) Basic Examples

Proposition 8.17

The Fourier transform of the Dirac mass at the origin is the constant distribution 1 :

$$\widehat{\delta_0} = T_1.$$

Proof:
For every $\varphi \in \mathcal{S}(\mathbb{R})$,

$$\widehat{\delta_0}(\varphi) = \delta_0(\widehat{\varphi}) = \widehat{\varphi}(0) = \int_{-\infty}^{\infty} \varphi(x)\,dx.$$

But this is exactly the action of the distribution associated with the constant function 1 :

$$T_1(\varphi) = \int_{-\infty}^{\infty} \varphi(x)\,dx.$$

Hence

$$\widehat{\delta_0} = T_1.$$

■

Proposition 8.18

The Fourier transform of the constant distribution 1 is the Dirac mass at the origin :

$$\widehat{T_1} = \delta_0.$$

Proof:
For every $\varphi \in \mathcal{S}(\mathbb{R})$,

$$\widehat{T_1}(\varphi) = T_1(\widehat{\varphi}) = \int_{-\infty}^{\infty} \widehat{\varphi}(x)\,dx.$$

By the Fourier inversion formula on $\mathcal{S}(\mathbb{R})$,

$$\int_{-\infty}^{\infty} \widehat{\varphi}(x)\,dx = \varphi(0).$$

Thus

$$\widehat{T_1}(\varphi) = \delta_0(\varphi),$$

and therefore

$$\widehat{T_1} = \delta_0.$$

■

9) Differentiation and Fourier Transform

Theorem 8.19

Let $T \in \mathcal{S}'(\mathbb{R})$. Then :

$$\widehat{T'}(\varphi) = 2\pi i\,\xi\,\widehat{T}$$

in the sense that

$$\widehat{T'} = 2\pi i\,\xi\,\widehat{T},$$

and

$$\widehat{xT} = \frac{i}{2\pi}\,(\widehat{T})'.$$

Proof:
Let $\varphi \in \mathcal{S}(\mathbb{R})$. Then

$$\widehat{T'}(\varphi) = T'(\widehat{\varphi}) = -T\big((\widehat{\varphi})'\big).$$

Now the differentiation formula on the Schwartz space gives

$$(\widehat{\varphi})'(x) = (\widehat{-2\pi i\xi\varphi(\xi)})(x).$$

Hence

$$\widehat{T'}(\varphi) = T\big(\widehat{2\pi i\xi\varphi}\big) = \widehat{T}(2\pi i\xi\varphi).$$

This is exactly the action of the distribution $2\pi i\xi\,\widehat{T}$, so

$$\widehat{T'} = 2\pi i\xi\,\widehat{T}.$$

Similarly,

$$\widehat{xT}(\varphi) = (xT)(\widehat{\varphi}) = T(x\widehat{\varphi}).$$

Using the identity

$$x\widehat{\varphi}(x) = \widehat{\left(\frac{i}{2\pi}\varphi'\right)}(x),$$

we obtain

$$\widehat{xT}(\varphi) = T\left(\widehat{\left(\frac{i}{2\pi}\varphi'\right)}\right) = \widehat{T}\left(\frac{i}{2\pi}\varphi'\right) = -\frac{i}{2\pi}(\widehat{T})'(\varphi).$$

Since

$$-\frac{i}{2\pi} = \frac{1}{2\pi i},$$

this gives

$$\widehat{xT} = \frac{i}{2\pi}(\widehat{T})'.$$

■

10) Conclusion

The space $\mathcal{S}'(\mathbb{R})$ of tempered distributions provides the natural setting in which :

1. smooth rapidly decreasing functions,
2. polynomially growing functions,
3. Dirac masses and their derivatives,
4. and the Fourier transform

can all be treated within the same unified framework.

In this setting, the Fourier transform becomes a perfectly symmetric automorphism, and differentiation is transformed into multiplication, exactly as in the classical case, but now for generalized functions as well.

IX - Convolution

Convolution plays a central role in Fourier analysis. It allows us to smooth functions and solve differential equations. We begin with its basic properties.

1) Definition and Basic Properties

Definition 9.1

Let $f, g \in L^1(\mathbb{R})$. The convolution of f and g is defined by

$$(f * g)(x) := \int_{-\infty}^{\infty} f(x-y) g(y)\, dy.$$

Proposition 9.2

If $f, g \in L^1(\mathbb{R})$, then :

1. $f * g$ is well-defined for almost every x ;
2. $f * g \in L^1(\mathbb{R})$;
3. $$\|f * g\|_{L^1(\mathbb{R})} \leqslant \|f\|_{L^1(\mathbb{R})} \|g\|_{L^1(\mathbb{R})}.$$

Proof:
We estimate

$$|f * g(x)| \leqslant \int_{-\infty}^{\infty} |f(x-y)||g(y)|\, dy.$$

Integrating in x and using Fubini's theorem,

$$\int_{-\infty}^{\infty} |f * g(x)|\, dx \leqslant \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x-y)||g(y)|\, dy\, dx.$$

Set $u = x - y$. Then

$$\int_{\mathbb{R}} |f(x-y)|\, dx = \int_{\mathbb{R}} |f(u)|\, du.$$

Hence

$$\|f * g\|_{L^1} \leqslant \left(\int_{\mathbb{R}} |f(u)|\, du \right) \left(\int_{\mathbb{R}} |g(y)|\, dy \right) = \|f\|_{L^1} \|g\|_{L^1}.$$

■

Proposition 9.3

Let $f, g \in L^1(\mathbb{R})$. Then :

1. $f * g = g * f$ (commutativity) ;
2. $(f * g) * h = f * (g * h)$ (associativity) ;
3. $f * (g + h) = f * g + f * h$ (linearity).

Proof:
All properties follow from changes of variables and Fubini's theorem. ■

2) Convolution and Fourier Transform

Theorem 9.4

Let $f, g \in \mathcal{S}(\mathbb{R})$. Then

$$\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi).$$

Proof:
We compute

$$\widehat{f * g}(\xi) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x-y)g(y)\, dy \right) e^{-2\pi i x\xi}\, dx.$$

By Fubini's theorem,

$$= \int_{\mathbb{R}} g(y) \left(\int_{\mathbb{R}} f(x-y)e^{-2\pi i x\xi}\, dx \right) dy.$$

Set $u = x - y$. Then

$$\int_{\mathbb{R}} f(x-y)e^{-2\pi i x\xi}\, dx = e^{-2\pi i y\xi} \int_{\mathbb{R}} f(u)e^{-2\pi i u\xi}\, du.$$

Hence

$$\widehat{f * g}(\xi) = \hat{f}(\xi) \int_{\mathbb{R}} g(y)e^{-2\pi i y\xi}\, dy = \hat{f}(\xi)\hat{g}(\xi).$$

■

Theorem 9.5

If $f, g \in \mathcal{S}(\mathbb{R})$, then

$$\widehat{fg} = \hat{f} * \hat{g}.$$

X - Applications

In this section, we illustrate some classical applications of the Fourier transform in harmonic analysis and physics. We begin with applications to differential equations, where the Fourier transform allows us to convert differential problems into algebraic ones. We then conclude with the proof of the Heisenberg uncertainty principle, which highlights a fundamental limitation in the simultaneous localization of a function and its Fourier transform.

1) Applications to Differential Equations

One of the most powerful features of the Fourier transform is that it converts differential operators into multiplication operators. This makes it possible to solve differential equations by purely algebraic manipulations in the frequency domain.

1- a) A First Example : A Linear Differential Equation

Theorem 10.1

Let $a \in \mathbb{C}$ and let $f \in \mathcal{S}(\mathbb{R})$. Suppose $u \in \mathcal{S}(\mathbb{R})$ satisfies

$$u'(x) + a\,u(x) = f(x).$$

Then its Fourier transform satisfies

$$(2\pi i\xi + a)\hat{u}(\xi) = \widehat{f}(\xi),$$

and therefore

$$\hat{u}(\xi) = \frac{\widehat{f}(\xi)}{2\pi i\xi + a},$$

whenever the denominator is nonzero.

Proof:
Applying the Fourier transform to both sides of

$$u'(x) + a\,u(x) = f(x),$$

we obtain

$$\widehat{u'}(\xi) + a\,\hat{u}(\xi) = \widehat{f}(\xi).$$

Using the differentiation formula,

$$2\pi i\xi\,\hat{u}(\xi) + a\,\hat{u}(\xi) = \widehat{f}(\xi),$$

that is,

$$(2\pi i\xi + a)\hat{u}(\xi) = \widehat{f}(\xi).$$

This yields the announced formula. ■
Thus, a first-order differential equation in the variable x becomes an algebraic equation in the variable ξ.

1- b) The Second Derivative and the Laplacian in Dimension One

Proposition 10.2

If $f \in \mathcal{S}(\mathbb{R})$, then

$$\widehat{f''}(\xi) = -(2\pi\xi)^2\widehat{f}(\xi).$$

Proof:
Using the formula for the Fourier transform of derivatives twice,

$$\widehat{f''}(\xi) = (2\pi i\xi)^2 \widehat{f}(\xi) = -4\pi^2\xi^2 \widehat{f}(\xi) = -(2\pi\xi)^2 \widehat{f}(\xi).$$

■

This identity shows that the second derivative, which is a differential operator, becomes multiplication by the nonnegative function $4\pi^2\xi^2$, up to a minus sign.

1- c) The One-Dimensional Heat Equation

We now consider the heat equation on $\mathbb{R}$:

$$\partial_t u(t,x) - \partial_x^2 u(t,x) = 0, \qquad t > 0,\ x \in \mathbb{R},$$

with initial condition

$$u(0,x) = f(x).$$

Theorem 10.3

Let $f \in \mathcal{S}(\mathbb{R})$. Then the unique Schwartz solution of the Cauchy problem

$$\begin{cases} \partial_t u(t,x) - \partial_x^2 u(t,x) = 0, \\ u(0,x) = f(x), \end{cases}$$

is given by

$$u(t,x) = \int_{-\infty}^{\infty} e^{-4\pi^2 t\xi^2} \widehat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

Proof:
Fix $t > 0$ and apply the Fourier transform with respect to the variable x. Denote

$$\widehat{u}(t,\xi) = \int_{-\infty}^{\infty} u(t,x) e^{-2\pi i x\xi}\, dx.$$

Then

$$\partial_t \widehat{u}(t,\xi) - \widehat{\partial_x^2 u}(t,\xi) = 0.$$

Using the previous proposition,

$$\partial_t \widehat{u}(t,\xi) + 4\pi^2\xi^2 \widehat{u}(t,\xi) = 0.$$

Thus, for each fixed ξ, $\widehat{u}(t,\xi)$ satisfies the ordinary differential equation

$$\frac{d}{dt}\widehat{u}(t,\xi) + 4\pi^2\xi^2 \widehat{u}(t,\xi) = 0,$$

with initial condition

$$\widehat{u}(0,\xi) = \widehat{f}(\xi).$$

Solving this equation, we obtain

$$\widehat{u}(t,\xi) = e^{-4\pi^2 t\xi^2} \widehat{f}(\xi).$$

Applying the inverse Fourier transform yields

$$u(t,x) = \int_{-\infty}^{\infty} e^{-4\pi^2 t\xi^2} \widehat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

■

1- d) The Heat Kernel

Definition 10.4

For $t > 0$, define the heat kernel
$$H_t(x) := \frac{1}{\sqrt{4\pi t}}\, e^{-\frac{x^2}{4t}}.$$

Proposition 10.5

For every $t > 0$,
$$\widehat{H_t}(\xi) = e^{-4\pi^2 t\xi^2}.$$

Proof:
Recall that
$$K_\delta(x) = \frac{1}{\sqrt{\delta}}\, e^{-\frac{\pi x^2}{\delta}}$$
has Fourier transform
$$\widehat{K_\delta}(\xi) = e^{-\pi\delta\xi^2}.$$
Choose
$$\delta = 4\pi t.$$
Then
$$K_{4\pi t}(x) = \frac{1}{\sqrt{4\pi t}}\, e^{-\frac{x^2}{4t}} = H_t(x),$$
and therefore
$$\widehat{H_t}(\xi) = e^{-\pi(4\pi t)\xi^2} = e^{-4\pi^2 t\xi^2}.$$

■

Theorem 10.6

Let $f \in \mathcal{S}(\mathbb{R})$. The solution of the heat equation with initial data f can be written as
$$u(t,x) = (H_t * f)(x) = \int_{-\infty}^{\infty} H_t(x-y) f(y)\, dy.$$

Proof:
By the convolution theorem,
$$\widehat{H_t * f}(\xi) = \widehat{H_t}(\xi)\widehat{f}(\xi) = e^{-4\pi^2 t\xi^2}\widehat{f}(\xi).$$
By the formula obtained in the previous theorem,
$$\widehat{u}(t,\xi) = e^{-4\pi^2 t\xi^2}\widehat{f}(\xi).$$
Hence
$$\widehat{u}(t,\xi) = \widehat{H_t * f}(\xi).$$
Since both functions belong to $\mathcal{S}(\mathbb{R})$, the Fourier inversion formula implies
$$u(t,x) = (H_t * f)(x).$$

■

1- e) Smoothing Effect of the Heat Kernel

Proposition 10.7

Let $f \in \mathcal{S}(\mathbb{R})$. Then for every $t > 0$, the function

$$u(t, \cdot) = H_t * f$$

belongs to $\mathcal{S}(\mathbb{R})$. Moreover, if $m \geqslant 0$, then

$$\partial_x^m u(t, x) = (\partial_x^m H_t) * f(x) = H_t * f^{(m)}(x).$$

Proof:
Since $H_t \in \mathcal{S}(\mathbb{R})$ for every $t > 0$ and $f \in \mathcal{S}(\mathbb{R})$, the convolution $H_t * f$ is again smooth, and derivatives may be passed under the integral sign. One gets

$$\partial_x^m (H_t * f) = (\partial_x^m H_t) * f.$$

The identity

$$(\partial_x^m H_t) * f = H_t * f^{(m)}$$

follows by repeated integration by parts. ■
Thus, even if the initial data are merely rough, the heat flow immediately smooths them out. This is one of the most fundamental regularizing effects in analysis.

1- f) The Inhomogeneous Equation

Consider now the inhomogeneous heat equation

$$\partial_t u(t, x) - \partial_x^2 u(t, x) = F(t, x), \qquad u(0, x) = f(x).$$

Theorem 10.8: Duhamel formula

If $f \in \mathcal{S}(\mathbb{R})$ and F is sufficiently regular, then the solution is given by

$$u(t, x) = (H_t * f)(x) + \int_0^t \big(H_{t-s} * F(s, \cdot)\big)(x)\, ds.$$

Proof:
Applying the Fourier transform in the variable x, we obtain

$$\partial_t \widehat{u}(t, \xi) + 4\pi^2 \xi^2 \widehat{u}(t, \xi) = \widehat{F}(t, \xi), \qquad \widehat{u}(0, \xi) = \widehat{f}(\xi).$$

For each fixed ξ, this is a first-order linear ODE in t. Using the integrating factor

$$e^{4\pi^2 t \xi^2},$$

we find

$$\widehat{u}(t, \xi) = e^{-4\pi^2 t \xi^2} \widehat{f}(\xi) + \int_0^t e^{-4\pi^2 (t-s) \xi^2} \widehat{F}(s, \xi)\, ds.$$

Taking the inverse Fourier transform, and using that

$$\widehat{H_{t-s}}(\xi) = e^{-4\pi^2 (t-s) \xi^2},$$

we obtain

$$u(t, x) = (H_t * f)(x) + \int_0^t (H_{t-s} * F(s, \cdot))(x)\, ds.$$

■

Conclusion

We have seen that :

1. differentiation in the variable x becomes multiplication by $2\pi i\xi$ after Fourier transformation ;
2. the heat equation is reduced to an ordinary differential equation in the frequency variable ;
3. its solution is expressed through convolution with the heat kernel.

This illustrates one of the central principles of harmonic analysis :

the Fourier transform turns differential equations into algebraic equations.

2) Poisson Summation Formula

We now present one of the most important identities in Fourier analysis, which connects discrete sums and continuous Fourier transforms.

Definition 10.9

Let $f \in \mathcal{S}(\mathbb{R})$. The 1–periodization of f (see Definition 3.1) is defined by

$$Pf(x) := \sum_{k\in\mathbb{Z}} f(x+k).$$

Proposition 10.10

If $f \in \mathcal{S}(\mathbb{R})$, then :
the series defining $Pf(x)$ converges absolutely and uniformly.

Proof:
Since $f \in \mathcal{S}(\mathbb{R})$, for every $N > 1$,

$$|f(x)| \leqslant \frac{C_N}{(1+|x|)^N}.$$

Thus

$$\sum_{k\in\mathbb{Z}} |f(x+k)| \leqslant C_N \sum_{k\in\mathbb{Z}} \frac{1}{(1+|x+k|)^N},$$

which converges uniformly in x. ∎

Proposition 10.11

The Fourier coefficients of Pf are given by

$$\widehat{Pf}(n) = \hat{f}(n), \qquad n \in \mathbb{Z}.$$

Proof:
Proceed exactly as in the periodization argument :

$$\widehat{Pf}(n) = \int_0^1 \sum_{k\in\mathbb{Z}} f(x+k)e^{-2\pi i n x}\,dx.$$

Interchanging sum and integral and using $u = x+k$, we obtain

$$\widehat{Pf}(n) = \int_{\mathbb{R}} f(u)e^{-2\pi i n u}\,du = \hat{f}(n).$$

∎

Theorem 10.12: Poisson summation formula

Let $f \in \mathcal{S}(\mathbb{R})$. Then for every $x \in \mathbb{R}$,

$$\sum_{k\in\mathbb{Z}} f(x+k) = \sum_{n\in\mathbb{Z}} \widehat{f}(n)e^{2\pi inx}.$$

Proof:
Since Pf is 1-periodic and smooth, it is equal to its Fourier series :

$$Pf(x) = \sum_{n\in\mathbb{Z}} \widehat{Pf}(n)e^{2\pi inx}.$$

Using $\widehat{Pf}(n) = \widehat{f}(n)$ gives the result. ■

Corollary 10.13

For $f \in \mathcal{S}(\mathbb{R})$,

$$\sum_{k\in\mathbb{Z}} f(k) = \sum_{n\in\mathbb{Z}} \widehat{f}(n).$$

3) Application : Gaussian Identity

Proposition 10.14

Let $f(x) = e^{-\pi x^2}$. Then

$$\sum_{k\in\mathbb{Z}} e^{-\pi(x+k)^2} = \sum_{n\in\mathbb{Z}} e^{-\pi n^2}e^{2\pi inx}.$$

This identity plays a fundamental role in number theory and the theory of modular forms.

This concludes the study of the Fourier transform on $\mathbb{R}$.

4) Application : Heisenberg Uncertainty Principle

The uncertainty principle expresses the fact that a function and its Fourier transform cannot both be highly concentrated. In other words, a signal cannot be sharply localized both in space and in frequency.

Theorem 10.15: Heisenberg uncertainty principle

Let $f \in L^2(\mathbb{R})$ be such that $xf \in L^2(\mathbb{R})$ and $f' \in L^2(\mathbb{R})$. Then

$$\left(\int_{\mathbb{R}} x^2|f(x)|^2\,dx\right)\left(\int_{\mathbb{R}} \xi^2|\widehat{f}(\xi)|^2\,d\xi\right) \geqslant \frac{1}{16\pi^2}\|f\|_{L^2(\mathbb{R})}^4.$$

Proof:
From the differentiation property of the Fourier transform (see Proposition 5.6),

$$\widehat{f'}(\xi) = 2\pi i\xi\,\widehat{f}(\xi).$$

By Plancherel's theorem (Theorem 7.3), we obtain

$$\|f'\|_{L^2(\mathbb{R})}^2 = \|\widehat{f'}\|_{L^2(\mathbb{R})}^2 = 4\pi^2\int_{\mathbb{R}} \xi^2|\widehat{f}(\xi)|^2\,d\xi.$$

Thus,

$$\int_{\mathbb{R}} \xi^2 |\widehat{f}(\xi)|^2 \, d\xi = \frac{1}{4\pi^2} \|f'\|_{L^2}^2.$$

We compute

$$\int_{\mathbb{R}} |f(x)|^2 \, dx = \int_{\mathbb{R}} f(x)\overline{f(x)} \, dx.$$

Observe that

$$\frac{d}{dx}\big(x|f(x)|^2\big) = |f(x)|^2 + x\frac{d}{dx}|f(x)|^2.$$

But

$$\frac{d}{dx}|f(x)|^2 = f'(x)\overline{f(x)} + f(x)\overline{f'(x)}.$$

Hence

$$\frac{d}{dx}\big(x|f(x)|^2\big) = |f(x)|^2 + x\big(f'(x)\overline{f(x)} + f(x)\overline{f'(x)}\big).$$

Integrating over $\mathbb{R}$ and using the fact that $xf \in L^2$ implies $x|f(x)|^2 \to 0$ at infinity, we obtain

$$\int_{\mathbb{R}} |f(x)|^2 \, dx = -\int_{\mathbb{R}} x\big(f'(x)\overline{f(x)} + f(x)\overline{f'(x)}\big) \, dx.$$

Taking the real part, we get

$$\|f\|_{L^2}^2 = -2\,\Re \int_{\mathbb{R}} x f'(x)\overline{f(x)} \, dx.$$

Using Cauchy–Schwarz,

$$\left| \int_{\mathbb{R}} x f'(x)\overline{f(x)} \, dx \right| \leqslant \|xf\|_{L^2} \, \|f'\|_{L^2}.$$

Hence

$$\|f\|_{L^2}^2 \leqslant 2 \, \|xf\|_{L^2} \, \|f'\|_{L^2}.$$

Squaring the inequality gives

$$\|f\|_{L^2}^4 \leqslant 4 \, \|xf\|_{L^2}^2 \, \|f'\|_{L^2}^2.$$

Using Step 1,

$$\|f'\|_{L^2}^2 = 4\pi^2 \int \xi^2 |\widehat{f}(\xi)|^2 \, d\xi.$$

Therefore

$$\|f\|_{L^2}^4 \leqslant 16\pi^2 \left(\int x^2 |f(x)|^2 \, dx \right) \left(\int \xi^2 |\widehat{f}(\xi)|^2 \, d\xi \right).$$

Dividing by $16\pi^2$ yields the desired inequality. ∎

Remark 10.16

Equality in Theorem 10.15 holds for Gaussian functions. More precisely, if

$$f(x) = Ce^{-kx^2}, \qquad C \in \mathbb{C}, \quad k > 0,$$

then equality takes place in the Heisenberg uncertainty inequality.

Indeed, let us examine the proof of Theorem 10.15. There we obtained

$$\|f\|_{L^2(\mathbb{R})}^2 = -2\,\Re \int_{\mathbb{R}} x f'(x)\overline{f(x)}\,dx,$$

and then estimated

$$\|f\|_{L^2(\mathbb{R})}^2 \leqslant 2 \left| \int_{\mathbb{R}} x f'(x)\overline{f(x)}\,dx \right| \leqslant 2\,\|xf\|_{L^2(\mathbb{R})}\,\|f'\|_{L^2(\mathbb{R})}.$$

Hence equality in Theorem 10.15 can occur only if equality holds in the Cauchy–Schwarz inequality, that is, only if the functions xf and f' are linearly dependent. Thus there must exist a constant $\lambda \in \mathbb{C}$ such that

$$f'(x) = \lambda x f(x).$$

This is a first-order differential equation. On any interval where f does not vanish, we may divide by f and obtain

$$\frac{f'(x)}{f(x)} = \lambda x.$$

Integrating gives

$$\log f(x) = \frac{\lambda}{2}x^2 + \text{constant},$$

so that

$$f(x) = Ce^{\lambda x^2/2}.$$

Since $f \in L^2(\mathbb{R})$, the real part of λ must be negative. Writing

$$\lambda = -2k, \qquad k > 0,$$

we obtain

$$f(x) = Ce^{-kx^2}.$$

Conversely, for such a Gaussian one has

$$f'(x) = -2kxf(x),$$

so that f' is exactly proportional to xf, and therefore equality holds in the Cauchy–Schwarz step used in the proof of Theorem 10.15. Consequently, equality holds in the uncertainty inequality.

In particular, for the normalized Gaussian

$$g(x) = e^{-\pi x^2},$$

we have $\widehat{g} = g$ (see Theorem 5.10), and this function is the most classical extremizer of Heisenberg's inequality.

Chapter 2

The Fourier Transform on $\mathbb{R}^d$

In the previous chapter, we studied the Fourier transform on $\mathbb{R}$. We now extend the theory to the multidimensional setting $\mathbb{R}^d$. The main ideas remain the same, but the notation becomes richer because we must take into account the Euclidean structure and partial derivatives with respect to several variables.

I - Euclidean Structure and Multi-Index Notation

Definition 1.1

For $x, \xi \in \mathbb{R}^d$, we define the Euclidean scalar product by

$$x \cdot \xi = \sum_{j=1}^{d} x_j \xi_j,$$

and the Euclidean norm by

$$|x| = (x \cdot x)^{1/2} = \left(\sum_{j=1}^{d} x_j^2 \right)^{1/2}.$$

We denote by dx the Lebesgue measure on $\mathbb{R}^d$.

Definition 1.2

A multi-index is an element

$$\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d.$$

We write

$$|\alpha| = \alpha_1 + \cdots + \alpha_d,$$

and, for $x = (x_1, \dots, x_d)$,

$$x^\alpha = x_1^{\alpha_1} \cdots x_d^{\alpha_d}.$$

We also define

$$D^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_d}^{\alpha_d}.$$

This notation allows us to write higher order partial derivatives and polynomial weights in a compact form.

II - The Schwartz Space on $\mathbb{R}^d$

Definition 2.1

The Schwartz space $\mathcal{S}(\mathbb{R}^d)$ is the space of all functions

$$f \in C^\infty(\mathbb{R}^d)$$

such that, for all multi-indices $\alpha, \beta \in \mathbb{N}^d$,

$$p_{\alpha,\beta}(f) := \sup_{x\in\mathbb{R}^d} \left|x^\alpha D^\beta f(x)\right| < \infty.$$

Thus, a function in $\mathcal{S}(\mathbb{R}^d)$ is smooth and rapidly decreasing at infinity, together with all its derivatives.

Remark 2.2

The quantities

$$p_{\alpha,\beta}(f) = \sup_{x\in\mathbb{R}^d} \left|x^\alpha D^\beta f(x)\right|$$

are seminorms. They measure both the decay of f at infinity and the behavior of its derivatives.

Proposition 2.3

The Schwartz space $\mathcal{S}(\mathbb{R}^d)$ is a vector space. Moreover, it is stable under differentiation and multiplication by polynomials.

Proof:

Let $f, g \in \mathcal{S}(\mathbb{R}^d)$ and let $\lambda, \mu \in \mathbb{C}$. For all multi-indices α, β, we have

$$x^\alpha D^\beta(\lambda f + \mu g) = \lambda x^\alpha D^\beta f + \mu x^\alpha D^\beta g.$$

Therefore,

$$|x^\alpha D^\beta(\lambda f + \mu g)| \leqslant |\lambda|\,|x^\alpha D^\beta f| + |\mu|\,|x^\alpha D^\beta g|.$$

Since both terms on the right are bounded, we obtain

$$p_{\alpha,\beta}(\lambda f + \mu g) < \infty.$$

Hence $\lambda f + \mu g \in \mathcal{S}(\mathbb{R}^d)$.

Now let $f \in \mathcal{S}(\mathbb{R}^d)$ and let γ be a multi-index. We prove that $D^\gamma f \in \mathcal{S}(\mathbb{R}^d)$. Indeed,

$$x^\alpha D^\beta(D^\gamma f) = x^\alpha D^{\beta+\gamma} f,$$

which is bounded because $f \in \mathcal{S}(\mathbb{R}^d)$.

Finally, let P be a polynomial. We show that $Pf \in \mathcal{S}(\mathbb{R}^d)$. By Leibniz's formula,

$$D^\beta(Pf) = \sum_{\rho\leqslant\beta} \binom{\beta}{\rho} D^\rho P\, D^{\beta-\rho} f.$$

Multiplying by x^α, we get a finite sum of terms of the form

$$x^\alpha (D^\rho P)(x) D^{\beta-\rho} f(x).$$

Since $D^\rho P$ is again a polynomial, each such term is a finite linear combination of expressions of the form

$$x^\eta D^\sigma f(x),$$

which are bounded because $f \in \mathcal{S}(\mathbb{R}^d)$. Hence $Pf \in \mathcal{S}(\mathbb{R}^d)$. ■

III - Definition of the Fourier Transform

Definition 3.1

Let $f \in \mathcal{S}(\mathbb{R}^d)$. The Fourier transform of f is defined by

$$\widehat{f}(\xi) = \mathcal{F}f(\xi) := \int_{\mathbb{R}^d} f(x)e^{-2\pi i x\cdot\xi}\,dx, \qquad \xi \in \mathbb{R}^d.$$

Proposition 3.2

If $f \in \mathcal{S}(\mathbb{R}^d)$, then the integral defining $\widehat{f}(\xi)$ is absolutely convergent for every $\xi \in \mathbb{R}^d$.

Proof:
Since $f \in \mathcal{S}(\mathbb{R}^d)$, for every integer $N \geqslant 0$ there exists a constant $C_N > 0$ such that

$$|f(x)| \leqslant \frac{C_N}{(1+|x|)^N}.$$

Choose $N > d$. Then

$$(1+|x|)^{-N} \in L^1(\mathbb{R}^d).$$

Hence $f \in L^1(\mathbb{R}^d)$. Since

$$|e^{-2\pi i x\cdot\xi}| = 1,$$

we obtain

$$\int_{\mathbb{R}^d} |f(x)e^{-2\pi i x\cdot\xi}|\,dx = \int_{\mathbb{R}^d} |f(x)|\,dx < \infty.$$

Therefore the Fourier transform is well-defined. ■

IV - Elementary Properties

The following proposition gathers the fundamental properties of the Fourier transform, which generalize those established in dimension 1.

Proposition 4.1

Let $f, g \in \mathcal{S}(\mathbb{R}^d)$ and let $\lambda, \mu \in \mathbb{C}$. Then :

(1) **Linearity.**

$$\mathcal{F}(\lambda f + \mu g)(\xi) = \lambda\,\widehat{f}(\xi) + \mu\,\widehat{g}(\xi).$$

(2) **Translation.** For $a \in \mathbb{R}^d$,

$$\mathcal{F}\big(f(\cdot + a)\big)(\xi) = e^{2\pi i a\cdot\xi}\,\widehat{f}(\xi).$$

(3) **Modulation.** For $a \in \mathbb{R}^d$,

$$\mathcal{F}\big(e^{-2\pi i a\cdot x} f(x)\big)(\xi) = \widehat{f}(\xi + a).$$

(4) **Dilation.** For $\delta \neq 0$,

$$\mathcal{F}\big(f(\delta x)\big)(\xi) = \frac{1}{|\delta|^d}\widehat{f}\left(\frac{\xi}{\delta}\right).$$

(5) **Differentiation.** For every multi-index α,

$$\widehat{D^\alpha f}(\xi) = (2\pi i\xi)^\alpha\,\widehat{f}(\xi).$$

(6) **Multiplication by monomials.** For every multi-index α,

$$\widehat{x^\alpha f}(\xi) = \left(\frac{i}{2\pi}\right)^{|\alpha|} D_\xi^\alpha \widehat{f}(\xi).$$

Moreover, for every $g \in L^1(\mathbb{R}^d)$,

$$\|\widehat{g}\|_{L^\infty(\mathbb{R}^d)} \leqslant \|g\|_{L^1(\mathbb{R}^d)}.$$

Proof:
We prove each property in turn.

(1) Linearity. For $\lambda, \mu \in \mathbb{C}$,

$$\begin{aligned}
\mathcal{F}(\lambda f + \mu g)(\xi) &= \int_{\mathbb{R}^d} (\lambda f(x) + \mu g(x)) e^{-2\pi i x\cdot\xi}\, dx \\
&= \lambda \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\cdot\xi}\, dx + \mu \int_{\mathbb{R}^d} g(x) e^{-2\pi i x\cdot\xi}\, dx \\
&= \lambda\, \widehat{f}(\xi) + \mu\, \widehat{g}(\xi).
\end{aligned}$$

(2) Translation. Let $a \in \mathbb{R}^d$. Then

$$\mathcal{F}(f(\cdot + a))(\xi) = \int_{\mathbb{R}^d} f(x+a) e^{-2\pi i x\cdot\xi}\, dx.$$

Using the change of variables $y = x + a$, we obtain

$$\begin{aligned}
\mathcal{F}(f(\cdot + a))(\xi) &= \int_{\mathbb{R}^d} f(y) e^{-2\pi i (y-a)\cdot\xi}\, dy \\
&= e^{2\pi i a\cdot\xi} \widehat{f}(\xi).
\end{aligned}$$

(3) Modulation. Let $a \in \mathbb{R}^d$. Then

$$\begin{aligned}
\mathcal{F}(e^{-2\pi i a\cdot x} f(x))(\xi) &= \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\cdot(\xi + a)}\, dx \\
&= \widehat{f}(\xi + a).
\end{aligned}$$

(4) Dilation. Let $\delta \neq 0$. Then

$$\mathcal{F}(f(\delta x))(\xi) = \int_{\mathbb{R}^d} f(\delta x) e^{-2\pi i x\cdot\xi}\, dx.$$

Using the change of variables $y = \delta x$, so that $dx = |\delta|^{-d} dy$, we obtain

$$\begin{aligned}
\mathcal{F}(f(\delta x))(\xi) &= \frac{1}{|\delta|^d} \int_{\mathbb{R}^d} f(y) e^{-2\pi i (y/\delta)\cdot\xi}\, dy \\
&= \frac{1}{|\delta|^d} \widehat{f}\left(\frac{\xi}{\delta}\right).
\end{aligned}$$

(5) Differentiation. We first treat the case of one derivative. For $1 \leqslant j \leqslant d$,

$$\widehat{\partial_{x_j} f}(\xi) = \int_{\mathbb{R}^d} \partial_{x_j} f(x) e^{-2\pi i x\cdot\xi}\, dx.$$

Integrating by parts in the variable x_j, and using the rapid decay of f to eliminate boundary terms, we obtain

$$\widehat{\partial_{x_j} f}(\xi) = 2\pi i \xi_j \widehat{f}(\xi).$$

Iterating this identity yields

$$\widehat{D^\alpha f}(\xi) = (2\pi i \xi)^\alpha \widehat{f}(\xi).$$

(6) Multiplication by monomials. We first consider the case of $x_j f$. Differentiating under the integral sign gives

$$\begin{aligned}\partial_{\xi_j} \widehat{f}(\xi) &= \frac{d}{d\xi_j} \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\cdot\xi}\, dx \\ &= \int_{\mathbb{R}^d} f(x)(-2\pi i x_j) e^{-2\pi i x\cdot\xi}\, dx \\ &= -2\pi i\, \widehat{x_j f}(\xi).\end{aligned}$$

Hence

$$\widehat{x_j f}(\xi) = \frac{i}{2\pi} \partial_{\xi_j} \widehat{f}(\xi).$$

Iterating this identity gives

$$\widehat{x^\alpha f}(\xi) = \left(\frac{i}{2\pi}\right)^{|\alpha|} D_\xi^\alpha \widehat{f}(\xi).$$

Finally, if $g \in L^1(\mathbb{R}^d)$, then

$$|\widehat{g}(\xi)| = \left| \int_{\mathbb{R}^d} g(x) e^{-2\pi i x\cdot\xi}\, dx \right| \leqslant \int_{\mathbb{R}^d} |g(x)|\, dx = \|g\|_{L^1(\mathbb{R}^d)}.$$

Taking the supremum over $\xi \in \mathbb{R}^d$ yields

$$\|\widehat{g}\|_{L^\infty(\mathbb{R}^d)} \leqslant \|g\|_{L^1(\mathbb{R}^d)}.$$

■

V - The Fourier Transform Preserves the Schwartz Space

Theorem 5.1

If $f \in \mathcal{S}(\mathbb{R}^d)$, then

$$\widehat{f} \in \mathcal{S}(\mathbb{R}^d).$$

Proof:

Let α, β be multi-indices. We must prove that

$$\sup_{\xi\in\mathbb{R}^d} \left| \xi^\alpha D_\xi^\beta \widehat{f}(\xi) \right| < \infty.$$

First, by differentiation under the integral sign,

$$D_\xi^\beta \widehat{f}(\xi) = \mathcal{F}\big((-2\pi i x)^\beta f(x)\big)(\xi).$$

Set

$$g(x) = (-2\pi i x)^\beta f(x).$$

Since $f \in \mathcal{S}(\mathbb{R}^d)$ and multiplication by polynomials preserves the Schwartz space, we have

$$g \in \mathcal{S}(\mathbb{R}^d).$$

Using the identity

$$\widehat{D^\alpha g}(\xi) = (2\pi i \xi)^\alpha \widehat{g}(\xi),$$

we obtain

$$\xi^\alpha \widehat{g}(\xi) = \frac{1}{(2\pi i)^{|\alpha|}} \widehat{D^\alpha g}(\xi).$$

Therefore,

$$\xi^\alpha D_\xi^\beta \widehat{f}(\xi) = \frac{1}{(2\pi i)^{|\alpha|}} D^\alpha \big[\widehat{(-2\pi i x)^\beta} f(x) \big](\xi).$$

Define

$$H_{\alpha,\beta}(x) = \frac{1}{(2\pi i)^{|\alpha|}} D^\alpha \big[(-2\pi i x)^\beta f(x) \big].$$

Then

$$\xi^\alpha D_\xi^\beta \widehat{f}(\xi) = \widehat{H_{\alpha,\beta}}(\xi).$$

By Leibniz's formula, $H_{\alpha,\beta}$ is a finite linear combination of terms of the form

$$x^\rho D^\sigma f(x),$$

and each such function belongs to $L^1(\mathbb{R}^d)$ because $f \in \mathcal{S}(\mathbb{R}^d)$. Hence

$$H_{\alpha,\beta} \in L^1(\mathbb{R}^d).$$

Using the estimate

$$\|\widehat{h}\|_{L^\infty} \leqslant \|h\|_{L^1},$$

we obtain

$$\sup_{\xi \in \mathbb{R}^d} \left| \xi^\alpha D_\xi^\beta \widehat{f}(\xi) \right| = \|\widehat{H_{\alpha,\beta}}\|_{L^\infty} \leqslant \|H_{\alpha,\beta}\|_{L^1} < \infty.$$

Thus all Schwartz seminorms of $\widehat{f}$ are finite, and therefore

$$\widehat{f} \in \mathcal{S}(\mathbb{R}^d).$$

■

VI - The Gaussian Function and Its Fourier Transform

Proposition 6.1

Let

$$G(x) = e^{-\pi |x|^2}, \qquad x \in \mathbb{R}^d.$$

Then

$$\widehat{G}(\xi) = e^{-\pi |\xi|^2}, \qquad \xi \in \mathbb{R}^d.$$

Proof:

We write

$$G(x) = \prod_{j=1}^{d} e^{-\pi x_j^2}.$$

Since the Fourier transform in $\mathbb{R}^d$ factorizes into a product of one-dimensional transforms, we obtain

$$\widehat{G}(\xi) = \prod_{j=1}^{d} \widehat{e^{-\pi x_j^2}}(\xi_j).$$

From the one-dimensional result,

$$\widehat{e^{-\pi x^2}}(\xi) = e^{-\pi \xi^2},$$

we conclude

$$\widehat{G}(\xi) = e^{-\pi |\xi|^2}.$$

■

VII - Approximate Identity

Definition 7.1

For $\delta > 0$, define

$$K_\delta(x) = \delta^{-d/2} e^{-\pi |x|^2/\delta}.$$

Proposition 7.2

The family $(K_\delta)_{\delta>0}$ satisfies :

(1) $K_\delta \geqslant 0$,

(2) $\displaystyle\int_{\mathbb{R}^d} K_\delta(x)\, dx = 1$,

(3) for every $\eta > 0$,

$$\int_{|x|\geqslant\eta} K_\delta(x)\, dx \longrightarrow 0 \quad \text{as } \delta \to 0.$$

Proof:
(1) is obvious.
For (2), use the change of variables $x = \sqrt{\delta}\, y$:

$$\int_{\mathbb{R}^d} K_\delta(x)\, dx = \int_{\mathbb{R}^d} e^{-\pi|y|^2}\, dy = 1.$$

For (3), using the same change of variables,

$$\int_{|x|\geqslant\eta} K_\delta(x)\, dx = \int_{|y|\geqslant\eta/\sqrt{\delta}} e^{-\pi|y|^2}\, dy.$$

Set $A = \eta/\sqrt{\delta}$. As $\delta \to 0$, we have $A \to \infty$.
Since $e^{-\pi|y|^2} \in L^1(\mathbb{R}^d)$ and is rapidly decreasing,

$$\int_{|y|\geqslant A} e^{-\pi|y|^2}\, dy \longrightarrow 0.$$

■

VIII - Convolution and Approximation

Definition 8.1

For $f, g \in L^1(\mathbb{R}^d)$, define
$$(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\,dy.$$

Proposition 8.2

If $f \in C^0(\mathbb{R}^d)$ is bounded and uniformly continuous, then
$$f * K_\delta(x) \longrightarrow f(x) \quad \text{as } \delta \to 0.$$

Proof:
Write
$$f * K_\delta(x) - f(x) = \int (f(x-y) - f(x))K_\delta(y)\,dy.$$
Split the integral into $|y| \leqslant \eta$ and $|y| > \eta$.
Use uniform continuity for the first part and small mass of K_δ for the second part. ∎

IX - Fourier Inversion Formula

Theorem 9.1

If $f \in \mathcal{S}(\mathbb{R}^d)$, then
$$f(x) = \int_{\mathbb{R}^d} \widehat{f}(\xi)e^{2\pi i x\cdot\xi}\,d\xi.$$

Proof:
Let
$$I_\delta(x) = \int_{\mathbb{R}^d} \widehat{f}(\xi)e^{-\pi\delta|\xi|^2}e^{2\pi i x\cdot\xi}\,d\xi.$$
Since
$$e^{-\pi\delta|\xi|^2} = \widehat{K_\delta}(\xi),$$
we obtain
$$I_\delta(x) = (f * K_\delta)(x).$$
Hence
$$I_\delta(x) \to f(x).$$
On the other hand, dominated convergence gives
$$I_\delta(x) \to \int_{\mathbb{R}^d} \widehat{f}(\xi)e^{2\pi i x\cdot\xi}d\xi.$$
This proves the result. ∎

X - Convolution Theorem

Theorem 10.1

If $f, g \in \mathcal{S}(\mathbb{R}^d)$, then
$$\widehat{f * g} = \widehat{f}\,\widehat{g}.$$

Proof:
Use Fubini and the change of variables $u = x - y$. ■

XI - Plancherel Identity

Proposition 11.1

If $f, g \in \mathcal{S}(\mathbb{R}^d)$, then
$$\int f(x)\overline{g(x)}dx = \int \widehat{f}(\xi)\overline{\widehat{g}(\xi)}d\xi.$$

Proof:
Use inversion formula and Fubini. ■

Theorem 11.2: Plancherel theorem

The Fourier transform extends uniquely to a unitary operator
$$\mathcal{F} : L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d).$$

XII - Applications to Partial Differential Equations

One of the main strengths of the Fourier transform is that it converts differential operators into multiplication operators. This allows us to solve certain partial differential equations explicitly.

We begin with a fundamental property of the Fourier transform.

Proposition 12.1

Let $f \in \mathcal{S}(\mathbb{R}^d)$. Then
$$\widehat{\Delta f}(\xi) = -4\pi^2|\xi|^2\widehat{f}(\xi),$$
where
$$\Delta f = \sum_{j=1}^{d} \frac{\partial^2 f}{\partial x_j^2}.$$

Proof:
Using the differentiation property,
$$\widehat{\partial_{x_j} f}(\xi) = 2\pi i\xi_j\widehat{f}(\xi),$$
we obtain
$$\widehat{\partial^2_{x_j} f}(\xi) = (2\pi i\xi_j)^2\widehat{f}(\xi) = -4\pi^2\xi_j^2\widehat{f}(\xi).$$
Summing over $j = 1, \ldots, d$, we get
$$\widehat{\Delta f}(\xi) = -4\pi^2\sum_{j=1}^{d}\xi_j^2\widehat{f}(\xi) = -4\pi^2|\xi|^2\widehat{f}(\xi).$$
■

1) The Heat Equation

We consider the Cauchy problem for the heat equation :

$$\begin{cases} \partial_t u(t,x) - \Delta u(t,x) = 0, \quad t > 0,\ x \in \mathbb{R}^d, \\ u(0,x) = f(x). \end{cases}$$

Theorem 12.2

Let $f \in \mathcal{S}(\mathbb{R}^d)$. The unique solution is given by

$$u(t,x) = \int_{\mathbb{R}^d} \widehat{f}(\xi) e^{-4\pi^2 t|\xi|^2} e^{2\pi i x\cdot\xi}\, d\xi.$$

Proof:
Take the Fourier transform in the spatial variable :

$$\widehat{u}(t,\xi) = \int_{\mathbb{R}^d} u(t,x) e^{-2\pi i x\cdot\xi} dx.$$

Applying the Fourier transform to the equation gives

$$\partial_t \widehat{u}(t,\xi) - \widehat{\Delta u}(t,\xi) = 0.$$

Using the previous proposition,

$$\partial_t \widehat{u}(t,\xi) + 4\pi^2 |\xi|^2 \widehat{u}(t,\xi) = 0.$$

This is an ordinary differential equation in t :

$$\partial_t v + av = 0, \quad a = 4\pi^2|\xi|^2.$$

Its solution is

$$\widehat{u}(t,\xi) = \widehat{f}(\xi) e^{-4\pi^2 t|\xi|^2}.$$

Applying the inversion formula gives

$$u(t,x) = \int_{\mathbb{R}^d} \widehat{f}(\xi) e^{-4\pi^2 t|\xi|^2} e^{2\pi i x\cdot\xi}\, d\xi.$$

■

Remark 12.3

The solution can be written as a convolution :

$$u(t,x) = (f * G_t)(x),$$

where

$$G_t(x) = \frac{1}{(4\pi t)^{d/2}} e^{-\frac{|x|^2}{4t}}$$

is the heat kernel.

2) The Poisson Equation

We consider the equation

$$-\Delta u = f \quad \text{in } \mathbb{R}^d.$$

Proposition 12.4

If $f \in \mathcal{S}(\mathbb{R}^d)$, then formally

$$\widehat{u}(\xi) = \frac{\widehat{f}(\xi)}{4\pi^2|\xi|^2}.$$

Remark 12.5

This formula is singular at $\xi = 0$, which shows that solving the Poisson equation requires additional conditions (for example, decay or mean-zero assumptions).

3) The Wave Equation

Consider

$$\partial_t^2 u - \Delta u = 0.$$

Taking the Fourier transform in space gives

$$\partial_t^2 \widehat{u}(t,\xi) + 4\pi^2|\xi|^2 \widehat{u}(t,\xi) = 0.$$

This is a second-order ODE :

$$\widehat{u}(t,\xi) = A(\xi)\cos(2\pi|\xi|t) + B(\xi)\sin(2\pi|\xi|t).$$

The coefficients $A(\xi)$ and $B(\xi)$ are determined by the initial conditions.

Remark 12.6

This shows that the Fourier transform diagonalizes the Laplacian and reduces PDEs to ordinary differential equations.

XIII - Uncertainty Principles on $\mathbb{R}^d$

We now present a fundamental limitation : a function cannot be simultaneously well localized in both space and frequency. This phenomenon is quantified by the Heisenberg uncertainty principle in higher dimensions.

The proof follows the same strategy as in the one-dimensional case, with appropriate adaptations using the Euclidean structure and multi-index notation.

1) Heisenberg Uncertainty Principle

Theorem 13.1: Heisenberg uncertainty principle on $\mathbb{R}^d$

Let $f \in \mathcal{S}(\mathbb{R}^d)$ with $f \neq 0$. Then

$$\left(\int_{\mathbb{R}^d} |x|^2 |f(x)|^2\, dx\right)\left(\int_{\mathbb{R}^d} |\xi|^2 |\widehat{f}(\xi)|^2\, d\xi\right) \geqslant \frac{d^2}{16\pi^2}\|f\|_{L^2(\mathbb{R}^d)}^4.$$

Proof:
Since $f \in \mathcal{S}(\mathbb{R}^d)$, all integrations by parts below are justified. We begin with the identity

$$\int_{\mathbb{R}^d} |f(x)|^2\,dx = -\frac{2}{d}\,\mathrm{Re}\int_{\mathbb{R}^d} x\cdot\nabla f(x)\,\overline{f(x)}\,dx.$$

Indeed, since

$$\nabla(|f|^2) = 2\,\mathrm{Re}(\nabla f\,\overline{f}),$$

we have

$$\mathrm{div}\big(x|f(x)|^2\big) = d|f(x)|^2 + x\cdot\nabla(|f(x)|^2).$$

Integrating over $\mathbb{R}^d$ and using the rapid decay of f, the boundary term vanishes, hence

$$0 = d\int_{\mathbb{R}^d} |f(x)|^2\,dx + 2\,\mathrm{Re}\int_{\mathbb{R}^d} x\cdot\nabla f(x)\,\overline{f(x)}\,dx.$$

Therefore,

$$\int_{\mathbb{R}^d} |f(x)|^2\,dx \leqslant \frac{2}{d}\int_{\mathbb{R}^d} |x|\,|\nabla f(x)|\,|f(x)|\,dx.$$

By the Cauchy–Schwarz inequality,

$$\int_{\mathbb{R}^d} |f(x)|^2\,dx \leqslant \frac{2}{d}\left(\int_{\mathbb{R}^d} |x|^2|f(x)|^2\,dx\right)^{1/2}\left(\int_{\mathbb{R}^d} |\nabla f(x)|^2\,dx\right)^{1/2}.$$

Squaring both sides gives

$$\left(\int_{\mathbb{R}^d} |x|^2|f(x)|^2\,dx\right)\left(\int_{\mathbb{R}^d} |\nabla f(x)|^2\,dx\right) \geqslant \frac{d^2}{4}\left(\int_{\mathbb{R}^d} |f(x)|^2\,dx\right)^2.$$

By Plancherel's theorem and the differentiation formula,

$$\widehat{\partial_{x_j} f}(\xi) = 2\pi i\xi_j \widehat{f}(\xi),$$

we obtain

$$\int_{\mathbb{R}^d} |\nabla f(x)|^2\,dx = 4\pi^2\int_{\mathbb{R}^d} |\xi|^2|\widehat{f}(\xi)|^2\,d\xi.$$

Substituting this into the previous inequality gives

$$\left(\int_{\mathbb{R}^d} |x|^2|f(x)|^2\,dx\right)\left(\int_{\mathbb{R}^d} |\xi|^2|\widehat{f}(\xi)|^2\,d\xi\right) \geqslant \frac{d^2}{16\pi^2}\left(\int_{\mathbb{R}^d} |f(x)|^2\,dx\right)^2.$$

This completes the proof. ■

Remark 13.2

The inequality shows that if f is strongly concentrated near the origin, then $\widehat{f}$ must be spread out in frequency, and conversely. The constant depends on the normalization

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} f(x)e^{-2\pi i x\cdot\xi}\,dx.$$

2) Variance Form

The preceding theorem is centered at the origin. A more flexible version measures concentration around arbitrary points.

Theorem 13.3: Variance form

Let $f \in \mathcal{S}(\mathbb{R}^d)$ and $f \neq 0$. Then for all $a, b \in \mathbb{R}^d$,

$$\left(\int_{\mathbb{R}^d} |x-a|^2 |f(x)|^2\, dx\right)\left(\int_{\mathbb{R}^d} |\xi-b|^2 |\widehat{f}(\xi)|^2\, d\xi\right) \geqslant \frac{d^2}{16\pi^2}\|f\|_{L^2(\mathbb{R}^d)}^4.$$

Proof:
Define

$$g(x) = e^{-2\pi i b\cdot x} f(x+a).$$

Then

$$|g(x)| = |f(x+a)|.$$

By the translation and modulation properties of the Fourier transform,

$$\widehat{g}(\xi) = e^{2\pi i a\cdot(\xi+b)}\widehat{f}(\xi+b).$$

Hence

$$|\widehat{g}(\xi)| = |\widehat{f}(\xi+b)|.$$

Applying the previous Heisenberg inequality to g gives

$$\left(\int_{\mathbb{R}^d} |x|^2 |g(x)|^2\, dx\right)\left(\int_{\mathbb{R}^d} |\xi|^2 |\widehat{g}(\xi)|^2\, d\xi\right) \geqslant \frac{d^2}{16\pi^2}\|g\|_2^4.$$

Changing variables in the first integral gives

$$\int_{\mathbb{R}^d} |x|^2 |g(x)|^2\, dx = \int_{\mathbb{R}^d} |y-a|^2 |f(y)|^2\, dy.$$

Similarly,

$$\int_{\mathbb{R}^d} |\xi|^2 |\widehat{g}(\xi)|^2\, d\xi = \int_{\mathbb{R}^d} |\eta-b|^2 |\widehat{f}(\eta)|^2\, d\eta.$$

Finally, $\|g\|_2 = \|f\|_2$, and the result follows. ■

3) Equality Case

Remark 13.4

Equality in the Heisenberg uncertainty principle is attained by Gaussian functions. More precisely, functions of the form

$$f(x) = Ce^{-\pi\lambda|x-a|^2} e^{2\pi i b\cdot x}, \qquad \lambda > 0,\ a, b \in \mathbb{R}^d,$$

are extremizers. This confirms the special role of the Gaussian in Fourier analysis.